

Revision Letter: Measuring Database Unfairness via Dependency Quantification Under DP

MAIN CHANGES

We thank the reviewers for their detailed and insightful comments, which have been immensely helpful in the revision process. We have significantly revised our paper in terms of experiments, completeness, and readability.

We first provide a summary of the main changes in the revision below, then we list detailed responses to comments from each reviewer.

Color codes. Updates made in response to specific comments are color-coded in the paper as follows.

- Changes mentioned by multiple reviewers and generic improvements in presentation are in Purple.
- Reviewer 1 is in Orange
- Reviewer 2 is in Magenta
- Reviewer 3 is in Red

META-REVIEW

Thanks for your submission to PVLDB volume 19. The reviewers appreciate that the paper addresses an important problem and that the paper is generally well-written. However, several concerns have also been raised, such as the need to further strengthen the motivation regarding unfairness with real-world examples, clarify how the metrics can be applied in practice, discuss how to incorporate the continuity and progression properties into the current design that focuses on static databases, enhance the experimental section to complement with the claims, and so on.

Response. We have made significant changes throughout the paper based on the reviewer comments. We summarize them here:

- **New use case experiments:** We have added **Section 5.2** with two new practical use-case experiments, addressing R1-D1, R3-D1, and R3-D7, and showing possible usage of the measures. In particular, we show that the measures can be used as a preliminary step before running exploratory data queries to assess bias before drawing conclusions from the query results. Furthermore, they provide a complementary view to estimating unfairness via machine learning models, removing dependence on the model class, hyperparameters, or the prediction task for which the model was trained.
- **Relation to dynamic properties from previous work:** We have added a dedicated discussion in the paragraph titled **Relation to other suggested properties** at the end of Section 3, where we address the continuity and progression properties proposed in previous work [69], in response to R2-W1 and R2-D1. We show that a similar guarantee to the continuity property holds for our measures, which stems from their sensitivity analysis, and further show that progression does not generally hold.
- **Interpreting the values of the measures:** In response to R1-D4, we have added an explanation on interpreting

the values of the measures in the paragraph titled **Practical use of the measures** in Section 3.3. We explain how the values of the measures can be interpreted both in absolute terms and relative to the values for other criteria, as further illustrated by the detailed use-case experiments in **Section 5.2**.

- **Improvements and added discussions:** Addressing R1-D1, R1-D2, R3-D7 and R3-D8, we have strengthened the merit of this paper both at the end of the paragraph **Contributions** in Section 1 and in **Section 6**, stating that, to the best of our knowledge, there is no prior work that focuses on measuring dataset unfairness under differential privacy, as well as motivates the use of data-level unfairness as a metric itself. Additionally, addressing R3-D4, R3-D3 and R3-D5, respectively, we have added a dedicated summary of the experimental results in the paragraph titled **Summary of our findings** at the beginning of Section 5 and improved the experimental setup detailed in **Section 5.1** by increasing the timeout and refining the handling of undefined or missing values during preprocessing. Furthermore, in response to R2-D2, we have provided additional evidence that $\mathcal{U}_{\text{MI}}^{\text{TV D}}$ is a good proxy for $\mathcal{U}_{\text{MI}}$ in the paragraph **Detecting varying unfairness** in Section 5.2. In response to R3-D2, we have added further justification in **Section 1** for the design of $\mathcal{U}_{\text{R}}^{\text{S:AT}}$ and $\mathcal{U}_{\text{TC}}$, emphasizing that they introduce novel perspectives on measuring unfairness. Finally, addressing R3-D4 and R3-D5, we have clarified some performance issues of the algorithms in **Section 5.3** and **Section 7**.
- **Figure presentation:** We have improved the presentation by increasing the resolution for all the figures throughout the paper, as well as changing the figure legends to display the unfairness measures' symbols (and the corresponding algorithms' numbers, when relevant) for clarity, addressing R2-W3, R2-D3, and R3-D9. For instance, all of the above was applied to **Figure 3a**, **Figure 3b** and **Figure 4**.

REVIEWER #1

Comment R1-D1. I'm not sure I'm convinced that measuring unfairness of the data is an improvement over measuring fairness of trained models. Is there a reason this is easier to measure under DP than fairness metrics of the model itself? It's definitely possible to train fair models on unfair data - much of the work in fairness is dedicated to this task - so it seems like measuring the models directly would be a better idea.

Response. We have added two new practical use-cases for the measures. Paragraph **A complementary view to Machine Learning models** in Section 5.2 shows that differentially private data-level unfairness metrics are more 'stable' than model-based ones due to the fact that model bias can be the result of other parameters beside the data, such as model class, hyperparameters, etc. In the paragraph **Assessing dataset biases in data exploration** in

Section 5.2 we also demonstrate that measuring dataset unfairness directly can be beneficial for data exploration and data analysis, for instance for the purposes of estimating the trustworthiness of query responses. We have additionally listed the conclusions from these experiments in the paragraph **Summary of our findings** in Section 5. Furthermore, we have expanded **Section 6** with the limitations found in state-of-the-art approaches to unfairness measurement in prediction systems, highlighting that they focus on model behavior or causal interpretations rather than measuring the unfairness of the data itself under DP.

Comment R1-D2. In addition, prior work has used unfairness of the data as a kind of objective to actually repair the data in order to train fair classifiers (rather than using the unfairness of the data as a metric). I don't think it's obvious that this is an important metric for evaluating data, and I think it would be good if the paper could provide further motivation for this.

Response. We further motivate the use of data-level unfairness as a metric at the end of the paragraph **Contributions** in Section 1, based on the new use case experiments in **Section 5.2**. In general, all our measures can be used for data analysis, as stated in the response for R1-D1. In particular, our repair-based measure, $\mathcal{U}_R^{SAT}$, is grounded in prior work on inconsistency measures [69], and offers a more interpretable notion of unfairness than other measures, as we now also explicitly mention in **Section 1 below Figure 2 and below Table 1 in Section 3**.

Comment R1-D3. The experimental results are comprehensive and show that the proposed algorithms work well. Correlation between the proxies and ground truth measurements seem to be strong for all proposed approaches.

Response. Thank you, we appreciate the comment.

Comment R1-D4. I'm not sure how someone would apply the approach in practice, however. The correlations seem strong, but the scale of each metric appears to depend on the dataset (based on my reading of Figure 7). Is there a way one could decide if a dataset is fair or unfair without knowing more about the data? For example, is there a dataset-agnostic threshold you could set to target a chosen demographic parity gap? I think this is important for the applicability of the approach, since a "relative" measure of unfairness won't really help someone decide whether or not they want to use the data.

Response. As we state in the paragraph **Practical use of the measures** at the end of Section 3.3, all proposed measures satisfy monotonicity and have well-defined lower and upper bounds, which correspond to absolute fair and unfair datasets, respectively. Additionally, in many cases, the scale of unfairness can vary across datasets and criteria, and so the user may wish to assess the unfairness of a criterion compared to a baseline criterion, rather than rely on the range, as we now show in **Section 5.2**. In particular, the paragraph **A complementary view to Machine Learning models** in Section 5.2 shows that for the Adult dataset, the unfairness measures give extremely small values for the criterion race $\perp$ sex, as we would expect, and give larger values for the criterion sex $\perp$ income > 50K. However, for ML models we get the opposite result (see detailed results in Table 4).

REVIEWER #2

Comment R2-W1. The framework is designed for a static database and therefore does not consider the continuity and progression properties. See D1 for details.

Response. We address this comment in our response to R2-D1, where we cover the discussion that we added on the continuity and progression properties, referring to the paragraph titled **Relation to other suggested properties** at the end of Section 3.3. We explain that a continuity-like guarantee follows from the sensitivity analysis of the measures, while progression does not generally hold.

Comment R2-W2. More empirical study on the tightness or effectiveness of the proxy measure is needed. See D2 for details.

Response. We address this comment in our response to R2-D2, where we provide the details on additional empirical evaluation of the tightness and effectiveness of the proxy $\mathcal{U}_{MI}^{TVD}$ measure. In particular, we have added a synthetic experiment comparing $\mathcal{U}_{MI}^{TVD}$ and $\mathcal{U}_{MI}$ as unfairness gradually increases, as displayed in the new **Figure 5b** and discussed in **Section 3.1** and in a paragraph **Detecting varying unfairness** in Section 5.2. We have also referenced the conclusions from this experiment in the paragraph **Summary of our findings** in Section 5.

Comment R2-W3. There are some presentation issues which should be addressed. See D3 for details.

Response. We address this comment in detail in our response to R2-D3, where we elaborate on the presentation issues that have been resolved: we have improved the resolution of all figures, revised the legends, and added the corresponding measures' symbols to them (as well as the corresponding algorithms' numbers, when relevant).

Comment R2-D1. In Section 3, the authors claim that their framework is designed for a static database, and therefore they do not consider the continuity and progression properties. It would be interesting if the authors could discuss how incorporating these properties might affect the framework. Specifically, what challenges or additional complexities would arise in the proposed measures when these properties are taken into account? A more detailed discussion of these aspects would provide deeper insight into the problem setting and help readers better understand the limitations and applicability of the framework.

Response. We have added a dedicated discussion on the continuity and progression properties in the paragraph **Relation to other suggested properties** at the end of Section 3.3. We now mention that we can derive a property that resembles continuity directly from the sensitivity analysis of the measures, which bounds the change in the measure between neighboring databases. In contrast, we show via a counterexample that the progression property does not hold for the measures. Specifically, for the defined set of allowed tuple modifications, there exist datasets in which no single modification reduces the unfairness expressed by the proposed measures.

Comment R2-D2. In Section 3.1, the authors claim that $\mathcal{U}_{MI}^{TVD}$ is bounded by $\mathcal{U}_{MI}$, and in practice $\mathcal{U}_{MI}^{TVD}$ could approximate $\mathcal{U}_{MI}$

tightly. Could the authors provide empirical evidence on how the tightness or effectiveness of the proposed $\mathcal{U}_{\text{MI}}^{\text{TV}D}$ varies with different data distributions? Such results would help clarify the conditions under which the proxy measure is reliable and its practical scope of applicability.

Response. We have added a new experiment, in which we compute $\mathcal{U}_{\text{MI}}^{\text{TV}D}$ and $\mathcal{U}_{\text{MI}}$ on a synthetic dataset with increasing unfairness, without privacy considerations. We display the result in **Figure 5b** and detail the results in the paragraph **Detecting varying unfairness** in Section 5.2 (we also have a forward reference in **Section 3.1**). From the experiment, we conclude that $\mathcal{U}_{\text{MI}}^{\text{TV}D}$ tightly approximates $\mathcal{U}_{\text{MI}}$, as well as grows proportionally to $\mathcal{U}_{\text{MI}}$ as the unfairness in the synthetic dataset grows. In addition, we compare $\mathcal{U}_{\text{MI}}^{\text{TV}D}$ and $\mathcal{U}_{\text{MI}}$ across multiple real-world datasets in **the full version of the paper [95]**, observing consistent alignment between the two measures, as we also state at the end of **Section 3.1** and in the paragraph **Summary of our findings** in Section 5. Overall, across different data distributions, $\mathcal{U}_{\text{MI}}^{\text{TV}D}$ showed itself as a good proxy for $\mathcal{U}_{\text{MI}}$, and so we found no conditions under which it was unreliable or impractical.

Comment R2-D3. Some figure legends are missing, unclear, or inconsistent with the corresponding plots (e.g., Fig. 9 and Fig. 10 in [82]), which may lead to confusion or misunderstanding. In addition, the resolution of some figures is relatively low, making certain details difficult to read. Clearer legends and higher-quality figures would help readers better interpret the results.

Response. We have addressed this issue by improving all figures in the paper. In particular, we have increased their resolution to enhance readability, as in **Figure 3a**. We have also revised the legends to ensure they are complete, clear, and consistent with the corresponding plots, for example in **Figure 3b**. We have also added measures' symbols (and the corresponding algorithms' numbers, when relevant) to the legends to further improve interpretability, as is also illustrated in the figures above.

REVIEWER #3

Comment R3-D1. Without any real-world examples, use cases, and results motivating the problem statement, it seems the problem is artificially curated. The authors should support the claims on the limitations of the state of the art duly.

Response. We have added two new practical use-case experiments in **Section 5.2**, which motivate the problem statement. The first experiment (paragraph **Assessing dataset biases in data exploration**) shows how unfairness measures can be used prior to running exploratory queries, as a measure of trust in the queries' results. The second experiment (paragraph **A complementary view to Machine Learning models**) highlights the limitations of model-based fairness metrics, demonstrating their dependence on model class, hyperparameters and the prediction task that they were trained for.

Comment R3-D2. The justification of designing $\mathcal{U}_{\text{MI}}^{\text{TV}D}$ is clear. However, why was designing $\mathcal{U}_{\text{R}}^{\text{SAT}}$ and $\mathcal{U}_{\text{TC}}$ necessary, given several similar metrics exist in the literature?

Response. We have added the justification for the novelty of the $\mathcal{U}_{\text{R}}^{\text{SAT}}$ and $\mathcal{U}_{\text{TC}}$ measures at the end of **Section 1**. We clarify that, unlike mutual information-based metrics, these measures provide complementary perspectives on unfairness, capturing how far the data is from being perfectly fair, or whether a subset of tuples dominates unfairness, respectively.

Comment R3-D3. Experiments that took longer than 2 hours were not included in the results. Why is this? What is the practical implication of an experiment taking longer to complete?

Response. We have increased the timeout to 24 hours and updated **Section 5.1** to indicate that. All the figures throughout the paper have been updated according to the new timeout, but in practice the timeout has not been exceeded in any of the experiments.

Comment R3-D4. What are the key takeaways of the scalability plots (Fig 5)? It seems that Algorithm 2 does not scale well. The authors simply stated the observations without explaining them.

Response. We have added the explanation for the fact that Algorithm 2 does not scale well in **Section 5.3**. This is due to the fact that its runtime depends on the runtime of the MaxSAT solver, which is substantial even when using the chunking heuristic, discussed in Section 5.1. We have also included this explanation in the new paragraph **Summary of our findings** at the beginning of Section 5 and in **Section 7**.

Comment R3-D5. Why is there a jump in the run time for two (of the three) algorithms as the criteria count is increased from 2 to 3?

Response. Thank you very much for this comment! The jump in runtime for $\mathcal{U}_{\text{MI}}^{\text{TV}D}$ and $\mathcal{U}_{\text{TC}}$ as the criteria count increases from two to three occurred due to an error in preprocessing of the datasets: rows with undefined or missing values in one of the columns of the examined fairness criterion were removed, and so there was large variance between the number of rows for different criteria. We have revised the preprocessing step, such that rows with missing values are no longer removed, but the missing and undefined values are treated as a separate value. We have documented the change in the preprocessing in the paragraph **Datasets** in Section 5.1.

Nevertheless, the runtime jump persists for the Healthcare dataset when the criteria count is increased from two to three. Therefore, we have analyzed this result and added an explanation in **Section 5.3**, stating that this happens due to the choice of the criteria and the small size of the dataset.

Comment R3-D6. In Section 5.5, the authors do a deep dive into the three measures and analyze how certain parameters/conditions affect their performance. Effectively, the authors analyze their proposed approaches against each other by subjecting them to specific conditions. I'm not sure this adds much value to the paper.

Response. We agree with this comment and have moved the relevant experiments to the Appendix section, available in [95]. We have added a brief mention of them in the paragraph **Summary of our findings** in Section 5.

Comment R3-D7. The authors should have more results that show how the proposed approaches address the limitations of the state of the art. The experiments can be thematically similar to the

ones they use to motivate the problem, highlighting the limitation of the state of the art.

Response. We have added two new practical use-case experiments in the new **Section 5.2**. The first experiment (paragraph **Assessing dataset biases in data exploration**) demonstrates how unfairness measures can be used to assess bias in data analysis and exploration scenarios, in order to estimate the trustworthiness of query results. The second experiment (paragraph **A complementary view to Machine Learning models**) illustrates the variability of model-based fairness metrics, showing that the bias indicated by the models can be the result of other parameters beside the data, and that our unfairness measures serve as a complementary and stable option to assess data unfairness for arbitrary criteria that do not necessarily involve the feature that the model aims to predict. We have also detailed the conclusions from these experiments in the paragraph **Summary of our findings** in Section 5. Furthermore, we now review previous work on machine learning model fairness in **Section 6** and state in the revised **Contributions** paragraph in

Section 1 that, to the best of our knowledge, there is no prior work that focuses on directly measuring database unfairness under DP.

Comment R3-D8. The paper could benefit from a discussion section that distills the merit of the proposed work and outlines the key takeaways.

Response. We have added a discussion section that distills the merit of the proposed work and outlines the key takeaways in the paragraph **Contributions** in Section 1. In the discussion we state that the measures can be used prior to running exploratory queries to know if query results are reliable, and as a stable and reliable alternative to methods of unfairness estimation via machine learning models.

Comment R3-D9. It would also be useful to augment the legends with the corresponding symbols for the three measures.

Response. We have added the corresponding measures' symbols to the legends (and the corresponding algorithms' numbers, when relevant) of all the figures throughout the paper. For example, **Figure 3a**, **Figure 3b** and **Figure 4** show this in their legends.

Measuring Database Unfairness via Dependency Quantification Under Differential Privacy

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ABSTRACT

Differential privacy (DP) has become the de facto standard for protecting sensitive data, providing strong guarantees that published statistics or models reveal limited information about any individual. However, privacy noise and restricted data access make it increasingly difficult to assess the fairness and reliability of private datasets. In this paper, we propose a formal framework for quantifying data unfairness under DP. We identify three core desiderata for unfairness measures based on previous work: positivity, monotonicity, and DP computability. We further instantiate them through three complementary measures: (1) a mutual information-based measure with a total variation distance proxy suitable for DP, (2) a data-repair-based measure approximated via a reduction to weighted MaxSAT, and (3) a top- k tuple contribution measure that isolates the most influential records in fairness violations. We design privacy-preserving algorithms and analyze their sensitivity, accuracy, and efficiency. Extensive experiments on multiple real-world datasets demonstrate that our proposed measures faithfully approximate their non-private counterparts, effectively quantify bias under privacy constraints, and provide insights for data management.

1 INTRODUCTION

Differential privacy (DP) [26, 29] has become the leading standard for safeguarding sensitive information in data analysis and machine learning (ML). It gives formal guarantees that published statistics or trained models reveal only limited information about any individual, even in the presence of auxiliary knowledge. DP has been widely adopted in practice by major organizations [24, 30, 87] and governmental agencies [2, 15, 37], and has been applied to a variety of tasks including private query answering [25, 44, 46, 58, 62, 71], synthetic data generation [6, 16, 38, 63, 64, 91, 100], and private machine learning [1, 50, 78]. Despite its broad success, DP introduces a fundamental challenge: while it ensures privacy, the injected noise and limited data access make it difficult to evaluate the *quality* and *fairness* of private data. Users and analysts must often rely on privacy-protected datasets to make decisions or train models, without being able to assess whether these datasets are accurate, representative, or equitable. Furthermore, learning from noisy or biased data can yield inaccurate and discriminatory predictions [82, 85].

Fairness and bias in data are central concerns in data management and ML, since biased data can produce unfair outcomes even when algorithms are well-designed. Extensive research has focused on defining algorithmic fairness, proposing formal notions such as Demographic Parity [13] and Conditional Statistical Parity [19], among many others [7, 27, 48, 55]. These notions constrain model behavior to ensure equitable treatment across protected groups. However, such definitions focus solely on the model, neglecting the data which itself may contain bias or unfairness. If the dataset

encodes biased relationships between protected and outcome attributes, an algorithm may reproduce or amplify those disparities.

In this work, we shift the focus from algorithmic fairness to the *fairness of data*. A key observation underlying our work is that many algorithmic fairness definitions can be expressed as forms of *conditional independence* [83]. For example, Demographic Parity requires independence between the sensitive attribute and the model outcome, while Conditional Statistical Parity introduces a conditioning set of admissible attributes. Using this formulation, we reinterpret fairness as a property of the *data distribution* itself rather than only of the learned classifier. This is motivated by a common assumption from a prior work [14, 32, 83]: if the labels in a dataset reflect the outcomes a reasonable classifier would produce, then dataset fairness becomes a reliable proxy for assessing the fairness of a classifier trained on it. Moreover, using the data in other ways, such as getting summarized statistics, may lead to biased conclusions. Consequently, testing whether a dataset satisfies such conditional independence statements provides a natural and principled way to assess its fairness. This connection enables the adaptation of well-established algorithmic fairness notions into *database-level fairness criteria*, which serve as the foundation for the framework we propose.

Our goal is to develop a principled, quantitative framework for measuring data unfairness that remains meaningful even when sufficient noise is added to it to satisfy DP. Prior research on measuring data quality has largely examined logical inconsistencies in data through integrity constraints, both without [9, 69, 79] and, recently, with privacy guarantees [74]. Yet, despite the growing literature on fairness in algorithms, no existing framework directly measures the unfairness of the private data itself, even though such measurements could give a valuable signal for data equity when the data cannot be fully observed.

Here, we introduce *differentially private unfairness measures* that, given a database D and a set of fairness criteria $\mathcal{F}$, compute a numerical score $\mathcal{U}(\mathcal{F}, D)$, quantifying how far the data deviates from satisfying those criteria. We formalize three key desiderata that any unfairness measure should satisfy, following principles devised for inconsistency measures [69]. First, the *Positivity* property [43, 79] ensures that the measure is non-negative and equals zero if and only if D satisfies all fairness criteria. Second, the *Monotonicity* property [79] states that expanding the set of fairness criteria ($\mathcal{F}_1 \subseteq \mathcal{F}_2$) cannot reduce the measured unfairness. Third, the measure must be efficiently and accurately computable under DP, maintaining interpretability and utility despite the added noise.

We present three complementary measures grounded in probabilistic dependence. The first, $\mathcal{U}_{\text{MI}}^{\text{TVD}}$, uses mutual information (MI) [21] to quantify dependence between sensitive and outcome attributes. Because MI is highly sensitive and thus unsuitable for

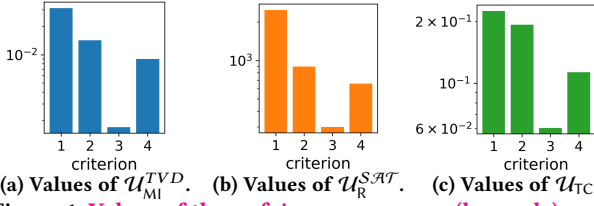


Figure 1: Values of the unfairness measures (log scale) on the Adult dataset for the four fairness criteria in Example 1.1.

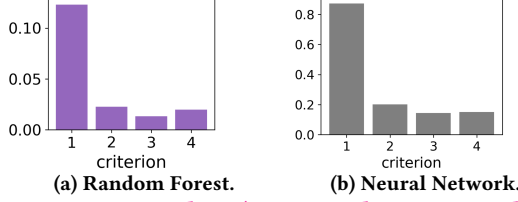


Figure 2: Fairness values (Demographic Parity and Conditional Statistical Parity gaps) of the two privately-trained ML models on the Adult dataset in Example 1.1.

DP [97], we develop a Total Variation Distance (TVD) proxy that closely approximates MI in both theory and practice. The second measure is based on data repair [18, 67, 83], quantifying the minimal number of tuple modifications required to make a dataset fair. Since this is computationally hard, we propose a proxy measure, $\mathcal{U}_R^{SAT}$, reducible to the weighted MaxSAT problem [93], following [83]. Finally, inspired by tuple contribution analysis [23, 65, 73, 96], we introduce a top- k contribution measure, $\mathcal{U}_{TC}$, that captures how much the most influential tuples contribute to fairness violations.

While $\mathcal{U}_{MI}^{TVD}$ approximates MI directly, $\mathcal{U}_R^{SAT}$ and $\mathcal{U}_{TC}$ introduce novel tuple-level unfairness notions tailored for DP, offering restricted sensitivity and greater interpretability than a distribution-level formula.

We prove that all three measures satisfy our desiderata, are close approximations to their originals, and exhibit low sensitivity relative to their range. We then design privacy-preserving algorithms for them that rely on the Laplace mechanism [28], and analyze their utility and complexity. Extensive experiments on five real-world datasets across diverse fairness settings confirm that our measures reliably capture comparative unfairness, remain faithful to their non-private baselines, and can be computed effectively under DP.

Example 1.1. Consider the Adult dataset¹ containing personal information of individuals in the US including whether their income is larger than 50K (we sample 10,000 tuples with 15 attributes). Figure 1 depicts the results of applying our three measures to four fairness criteria: (1) a person’s income should not depend on their years of education, (2) a person’s income should not depend on their sex, (3) a person’s income should not depend on their race, and (4) a person’s income should not depend on their sex, given that they work a certain number of hours per week (the criteria are phrased as independence statements in Table 2 in the sequel). We expect the first criterion to clearly be violated, as more education often leads to higher salaries, while the latter three criteria are natural and desired for most applications to avoid discriminatory conclusions and decision-making. Furthermore, it is known that

¹See the descriptions and links to the datasets in Section 5.1.

Adult has disparity in income for different sexes and races [89, 92]. Each measure was evaluated using our DP algorithms with a privacy budget of $\epsilon = 1$, with the experiment repeated ten times and the results averaged for each criterion. All the measures show a similar trend for criteria (1), (2), (3) and (4).

We additionally privately trained two models on the same sample using a budget of $\epsilon = 10$ and all available attributes to predict the outcome: (a) a DP version of RandomForest, which achieved 78.9% accuracy on average, and (b) a neural network with one hidden layer with 32 nodes and ReLU activation, trained with DP-SGD, which achieved 84.1% accuracy on average. Figure 2 shows the Demographic Parity gaps for criteria (1), (2) and (3), based on the classifier’s predictions: $|\max_{p \in P} Pr_D(O = 1 | P = p) - \min_{p \in P} Pr_D(O = 1 | P = p)|$ where $O = 1$ is a prediction of the income being bigger than 50K. Intuitively, this measures the disparity in positive outcomes for the most privileged group and the most discriminated one in a population consisting of more than two groups. The figure also shows the Conditional Statistical Parity gap for the conditional criterion (4), which is calculated as the expectation of the Demographic Parity over the conditioned attribute [80, 83]. We can see that the trend of our measures matches the trend of the ‘fairness measures’ commonly applied to ML models, indicating that the three measures are able to estimate the underlying unfairness based on the criteria, and that they might also be useful in predicting whether private data can be effectively used for various applications.

Contributions. This is the first work that provides a practical framework for quantifying private data unfairness through three different notions, specifically adapted for DP. In summary, the paper makes the following contributions:

- (1) We introduce a **formal framework for measuring data unfairness**, focusing on quantifying bias directly at the data level rather than at the algorithmic level, and defining general desiderata of *positivity*, *monotonicity*, and *DP computability*.
- (2) We propose **three concrete unfairness measures**, grounded respectively in (a) *mutual information* and its *total variation distance* proxy suitable for DP, $\mathcal{U}_{MI}^{TVD}$, (b) a *data repair* inspired measure, $\mathcal{U}_R^{SAT}$, approximated via a reduction to *weighted MaxSAT*, and (c) *tuple-level contribution* measure, $\mathcal{U}_{TC}$, identifying the top- k most influential records in fairness violations.
- (3) We design **DP algorithms** for computing the measures, and give formal analysis of their guarantees, error, and complexity.
- (4) We conduct an **extensive experimental evaluation** on five real-world datasets, demonstrating that our proposed measures **can be used in data analysis scenarios, demonstrate that they provide a complementary, stable, and reliable alternative to estimating unfairness via machine learning models, and that they follow trends of increasing data unfairness**. Finally, we show that the measures faithfully approximate their non-private counterparts and effectively quantify data unfairness under DP, and scale to large datasets.

2 PRELIMINARIES

We now give the necessary background for the paper, including basic database and probabilistic notions, fairness concepts, and essential definitions for our use of differential privacy.

2.1 Databases and Probabilities

We consider a single-relation schema $\mathcal{A} = (A_1, \dots, A_m)$, which is a vector of distinct attribute names A_i , each associated with a domain $\text{Dom}(A_i)$ of values. A database D over $\mathcal{A}$ is associated with a set $\text{tids}(D)$ of *tuple identifiers*, and it maps every identifier $i \in \text{tids}(D)$ to a tuple $D[i] = (a_1, \dots, a_m)$ in $A_1 \times \dots \times A_m$. We denote the size of the database by $|D| = n$. A tuple $t_i \in D$ has a specific value in each of its attributes, denoted by $t_i[A] = a$ where $a \in \text{Dom}(A)$. We consider bag semantics where t_i, t_j may share the same values across all attributes except their identifiers.

The computation of empirical probabilistic quantities in databases is often practically performed by measuring database statistics [31, 47, 99]. Let D be a dataset of size n , the *empirical marginal* of a value $A = a$ is defined as follows: $\Pr_D(A = a) = \frac{|\{t \mid t[A] = a\}|}{n}$. Similarly, the *conditional marginal* of $A = a$ conditioned on $C = c$ is

$$\Pr_D(A = a \mid C = c) = \frac{|\{t \mid t[A] = a, t[C] = c\}|}{|\{t \mid t[C] = c\}|}$$

When a set of attributes P equals a set of corresponding values p , we abuse notation and write $P = p$ when clear from context.

Employing these notations, we review the definition of mutual information, which is commonly used to quantify dependencies between variables and will be the basis for one of our measures.

Definition 2.1 (Mutual Information (MI) [21]). *The conditional mutual information between two attributes A and B conditioned on attribute C over their domain in a database D is defined as follows.*

$$MI_D(A, B \mid C) = \sum_{c \in \text{Dom}(C)} \Pr_D(C = c) \sum_{a \in \text{Dom}(A)} \sum_{b \in \text{Dom}(B)} \Pr_D(A = a, B = b \mid C = c) \log \left(\frac{\Pr_D(A = a, B = b \mid C = c)}{\Pr_D(A = a \mid C = c) \Pr_D(B = b \mid C = c)} \right)$$

where C may be absent for the unconditional case.

We say that two attributes $A, B \in \mathcal{A}$ in a database D are independent if $\Pr_D(A = a, B = b) = \Pr_D(A = a) \cdot \Pr_D(B = b)$ for all $a \in \text{Dom}(A)$ and $b \in \text{Dom}(B)$, or, equivalently, if $MI_D(A, B) = 0$, and we use the standard notation of independence $A \perp B$.

The notion of database probabilities can be used to express associational fairness definitions, as we discuss next.

2.2 Fairness as Independence Statements

Multiple fairness definitions that have been considered by previous work can be described as (conditional) independence statements. Demographic Parity [13] measures whether the favorable prediction $\hat{O}$ of an outcome (e.g., an income will be above a threshold) is affected by which demographic group the person belongs to, as captured by a protected attribute $P \in \mathcal{A}$ (e.g., Race, Sex). This statement can also be expressed by $P \perp \hat{O}$. Conditional Statistical Parity [19] further relaxes this definition by allowing certain admissible attributes $A \in \mathcal{A}$ (e.g., education level, work experience) to explain the unfairness, and can be written as $P \perp \hat{O} \mid A$. Equality of Opportunity [48] defines fairness by stating that for each case of the ground truth (e.g., the loan is indeed approved), as captured by the outcome attribute $O \in \mathcal{A}$, the probability of a favorable outcome should be equal across all demographic groups, i.e., $P \perp \hat{O} \mid O$. Predictive Parity [17] measures whether predicted positives are equally accurate across demographic groups, or equivalently, whether the

ground truth is independent of the protected attribute given the predicted outcome, i.e., $P \perp O \mid \hat{O}$.

We will simplify the discussion for the purpose of defining database fairness and distinguish between a protected attribute $P \in \mathcal{A}$, an outcome attribute $O \in \mathcal{A}$, and an admissible attribute $A \in \mathcal{A}$. Thus, we can generally define a *database fairness criterion* as follows.

Definition 2.2 (Fairness criterion). *Given a schema $\mathcal{A}$ with a protected attribute $P \in \mathcal{A}$, outcome attribute $O \in \mathcal{A}$, and admissible attribute $A \in \mathcal{A}$, a fairness criterion is an independence statement $P \perp O \mid A$, where A may be absent.*

A set of multiple fairness criteria will be denoted by $\mathcal{F}$.

We now consider the fact that the databases we wish to analyze are protected with differential privacy.

2.3 Differential Privacy

We next give the necessary definitions for DP.

Definition 2.3 (Neighboring Databases). *Two databases D, D' with the same schema are called neighboring if they have the same size and differ in exactly one tuple, denoted by $D' \sim D$.*

We often use the notion of neighboring databases to distinguish the impact of any particular individual's input on the output of a function. We likewise measure the maximum change in any function due to the replacement of a single tuple in the database, often calling it the sensitivity of the function.

Definition 2.4 (Sensitivity). *Given a function f the sensitivity of f is $\max_{D' \sim D} |f(D) - f(D')|$ and is denoted by Δ_f .*

Differential privacy (DP) [26] protects the private information of individuals in the data by ensuring similar results for similar databases with high probability.

Definition 2.5 (Differential Privacy [26]). *Given a privacy budget $\epsilon > 0$, an algorithm $\mathcal{M}$ is said to satisfy ϵ -DP if for all $S \subseteq \text{Range}(\mathcal{M})$ and for all $D \sim D'$,*

$$\Pr(\mathcal{M}(D) \in S) \leq e^\epsilon \Pr(\mathcal{M}(D') \in S)$$

The Laplace mechanism [28] allows us to enforce DP by adding calibrated noise to algorithm results and is a common building block in DP mechanisms.

Definition 2.6 (Laplace Mechanism [28]). *Given a database D , a function $f : \mathcal{D} \rightarrow \mathbb{R}$, and a privacy budget ϵ , the Laplace mechanism $\mathcal{M}_L$ returns $f(D) + v_q$, where $v_q \sim \text{Lap}(\Delta_f/\epsilon)$.*

THEOREM 2.7 (PRIVACY OF THE LAPLACE MECHANISM [29]). *Let $f : \mathcal{D} \rightarrow \mathbb{R}$ be a query with the sensitivity Δ_f , and fix $\epsilon > 0$. The Laplace mechanism $\mathcal{M}_L(D) = f(D) + v$, with $v \sim \text{Lap}(\Delta_f/\epsilon)$, satisfies ϵ -differential privacy.*

3 PRIVATELY EVALUATING UNFAIRNESS

We formulate the problem of privately evaluating the unfairness of a database, starting with a discussion about measure requirements.

Measure desiderata. We define several requirements for private unfairness measures. The first two are inspired by the properties outlined for inconsistency measures in the context of logical integrity constraints [69], and the third one is specific to the DP

Table 1: Properties of the discussed unfairness measures along with their computation and utility costs. Green rows denote the chosen measures and red rows denote measures that are unsuitable either due to relatively high sensitivity ($\mathcal{U}_{\text{MI}}$), unsatisfactory faithfulness to the original measure ($\mathcal{U}_{\text{MI}}^{\text{Bayes}}$), or high computation costs ($\mathcal{U}_{\text{R}}$). All measures satisfy the positivity and monotonicity properties outlined in Section 3, except $\mathcal{U}_{\text{MI}}^{\text{Bayes}}$ that violates positivity.

Motivation	Measure	Range for F	$\Delta_{\mathcal{U}}$ for F	$\Delta_{\mathcal{U}}$ for $\mathcal{F}$	Complexity	Error Bound
Attribute correlation	$\mathcal{U}_{\text{MI}}$	$[0, \infty)$	$O\left(\frac{\log n}{n}\right)$	-	-	-
	$\mathcal{U}_{\text{MI}}^{\text{Bayes}}$	$[-1, 0]$	$O\left(\frac{1}{n}\right)$	-	-	-
	$\mathcal{U}_{\text{MI}}^{\text{TV D}}$	$[0, 2]$	$O\left(\frac{1}{n}\right)$	$O\left(\frac{ \mathcal{F} }{n}\right)$	$O(\mathcal{F} n)$	$\frac{16 \mathcal{F} }{n\varepsilon}$
Distance to a fair database	$\mathcal{U}_{\text{R}}$	$[0, n]$	1	-	Exp. in n	-
	$\mathcal{U}_{\text{R}}^{\text{SAT}}$	$[0, n]$	2	$2 \mathcal{F} $	$O(\mathcal{F} (n^4 + \text{SAT}))$	$\frac{2 \mathcal{F} }{\varepsilon}$
Contribution of the top- k tuples to unfairness	$\mathcal{U}_{\text{TC}}$	$[0, \min\{\frac{k}{4}, 2\}]$	$O\left(\frac{k}{n}\right)$	$O\left(\frac{ \mathcal{F} k}{n}\right)$	$O(\mathcal{F} n \log n)$	$\frac{7k \mathcal{F} }{n\varepsilon}$

setting. We aim to design an unfairness measure that takes a private database D , a set of fairness criteria $\mathcal{F}$, and a privacy budget ε , and returns a non-negative number, i.e., $\mathcal{U}(\mathcal{F}, D) \in [0, \infty)$. We want $\mathcal{U}$ to have the following desiderata:

- (1) **Positivity:** $\mathcal{U}(\mathcal{F}, D) \geq 0$ and $\mathcal{U}(\mathcal{F}, D) = 0$ iff D satisfies F for every $F \in \mathcal{F}$.
- (2) **Monotonicity:** $\mathcal{U}(\mathcal{F}, D) \leq \mathcal{U}(\mathcal{F}', D)$ if $\mathcal{F} \subseteq \mathcal{F}'$.
- (3) **Computability under DP:** We can efficiently compute $\mathcal{U}$ with an ε -DP algorithm with relatively small error.

The first two properties have already been suggested for inconsistency measures. Positivity was proposed as a fundamental property in [43, 79] and as an axiom in [70]. Monotonicity was proposed in different variations by previous work. Parisi and Grant [79] proposed measure monotonicity for database containment, i.e., $D \subseteq D'$ implies $\mathcal{U}(\mathcal{F}, D) \leq \mathcal{U}(\mathcal{F}, D')$, yet, in our case, database containment does not guarantee an increase of fairness, and in fact, may decrease it. Livshits et al. [69] suggested monotonicity for logical implication of integrity constraints, i.e., if $\Sigma \models \Sigma'$ for two sets of integrity constraints Σ, Σ' , then $\mathcal{U}(\Sigma', D) \leq \mathcal{U}(\Sigma, D)$. Since comparing fairness criteria (see Definition 2.2) through logical implication is not possible, we define a new notion of monotonicity for fairness criteria set containment. Intuitively, when more fairness criteria are used, the requirements from the database are stricter and, therefore, the value of the unfairness measure should only increase. The third property will be shown in two parts. First, showing that our measures have low sensitivity (Section 3) and then providing a DP algorithm for computing them (Section 4).

In the sequel, we will first define the measures for a single fairness criterion, $\mathcal{U}(F, D)$, and then extend them to a set of criteria as a sum $\mathcal{U}(\mathcal{F}, D) = \sum_{F \in \mathcal{F}} \mathcal{U}(F, D)$. We will then prove the properties for this extension. As we will see, measures and techniques from previous work in the non-private setting can be unsuitable for computation under DP, motivating our third desideratum and leading us to search for proxies or replacements.

The other two properties listed in [69] are tailored for dynamic settings and are discussed in detail at the end of this section.

The remainder of Section 3 presents our three measures, outlining their motivation and theoretical properties. We begin with

$\mathcal{U}_{\text{MI}}^{\text{TV D}}$, which approximates mutual information between the attributes appearing in the fairness criteria (Section 3.1). We then introduce $\mathcal{U}_{\text{R}}^{\text{SAT}}$, a proxy for the minimal number of tuple insertions or deletions required to transform the dataset into one that satisfies the criteria (Section 3.2). Finally, we describe $\mathcal{U}_{\text{TC}}$, which reports the cumulative contribution of the top- k most influential tuples to the fairness criteria violation (Section 3.3). Unlike $\mathcal{U}_{\text{MI}}^{\text{TV D}}$, which approximates MI at the distribution level, $\mathcal{U}_{\text{R}}^{\text{SAT}}$ and $\mathcal{U}_{\text{TC}}$ are grounded in tuple-level notions of unfairness, compatible with DP, and greater interpretability. Table 1 provides a consolidated overview of the properties of all three measures.

3.1 Unfairness as Mutual Information

A standard way to measure dependence between variables is computing the mutual information (MI) between them [59, 72, 97]. Thus, for a fairness criterion $P \perp\!\!\!\perp O \mid A$, we can measure the MI between P and O given A to obtain our first measure notion. We abuse the notation from Definition 3.1 and denote the sum of (conditional) mutual information between the attributes in the fairness criteria contained in $\mathcal{F}$ by $MI_D(\mathcal{F})$, i.e., $MI_D(\mathcal{F}) = \sum_{F \in \mathcal{F}} MI_D(F)$.

Definition 3.1 (Mutual Information Unfairness). *Given a database D and a fairness criterion of the form $F = P \perp\!\!\!\perp O \mid A$ where A can be an empty set, the mutual information unfairness measure is defined as $\mathcal{U}_{\text{MI}}(F, D) = MI_D(P, O \mid A)$.*

Previous work [97] has already shown that $\mathcal{U}_{\text{MI}}$ is not suitable as an accurate measure in the DP setting due to its relatively high sensitivity of $O(\log n/n)$. Such sensitivity means that Laplace noise of scale proportionate to $\log n/(\varepsilon n)$ should be added to achieve ε -differential privacy. However, the range of $\mathcal{U}_{\text{MI}}$ is $[0, \infty)$, and the closer we get to F being satisfied (that is, the more independent the attributes in the F are), the smaller $\mathcal{U}_{\text{MI}}$ becomes. Thus, when F nearly holds and $\mathcal{U}_{\text{MI}}$ is close to 0, the added Laplace noise can severely distort the original $\mathcal{U}_{\text{MI}}$ value, making it unusable. The range of $\mathcal{U}_{\text{MI}}$ also poses a problem. Since it is unbounded, it is difficult to interpret its values and distinguish which values indicate low dependence score between the sensitive and outcome attributes, and which values indicate high dependence score. Therefore, we need to find an alternative with lower sensitivity.

To mitigate these issues, one can use a proxy function for $\mathcal{U}_{\text{MI}}$ that has low sensitivity and a bounded range. A possible proxy function, presented in [97] for a single fairness criterion, is $\mathcal{U}_{\text{MI}}^{\text{Bayes}}(F, D) = -\frac{1}{2} \min_{P \in \mathcal{P}^\circ} \|Pr^\circ[P, O \mid A] - Pr[P, O \mid A]\|$, where $Pr^\circ[P, O \mid A]$ is the maximum joint distribution defined as the one that maximizes the mutual information between P and O , given A . Zhang et al. [97] also showed that the sensitivity of $\mathcal{U}_{\text{MI}}^{\text{Bayes}}$ is $\frac{1}{n}$ for an unconditional fairness criterion, and is therefore suitable for DP.

However, $\mathcal{U}_{\text{MI}}^{\text{Bayes}}$ lacks the desired positivity property. Furthermore, the relation between $\mathcal{U}_{\text{MI}}$ and $\mathcal{U}_{\text{MI}}^{\text{Bayes}}$ can be difficult to visually comprehend. In the full version [95] we demonstrate the faithfulness between $\mathcal{U}_{\text{MI}}$ and $\mathcal{U}_{\text{MI}}^{\text{Bayes}}$ for the Adult, Stackoverflow survey, and Compas datasets (see Example 1.1) with fairness criteria detailed in Table 2 in the sequel.

This faithfulness gap motivates us to define a novel proxy function for $\mathcal{U}_{\text{MI}}$ that has low sensitivity and high correlation with $\mathcal{U}_{\text{MI}}$ both theoretically and in practice. This proxy measure is based on the concepts of Total Variation Distance (TVD), that is defined as $\text{TVD}(P, Q) = \frac{1}{2} \sum_x |P(x) - Q(x)|$ for two distributions P and Q .

Definition 3.2 ($\mathcal{U}_{\text{MI}}^{\text{TVD}}$ as a proxy for $\mathcal{U}_{\text{MI}}$). *Given a database D and a fairness criterion of the form $F = P \perp O \mid A$ where A can be an empty set, the $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ proxy for $\mathcal{U}_{\text{MI}}$ is defined as:*

$$\mathcal{U}_{\text{MI}}^{\text{TVD}}(P \perp O \mid A, D) = 2 \cdot (\text{TVD}(Pr_D(P, O \mid A), Pr_D(P \mid A)Pr_D(O \mid A)))^2$$

We first show that $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ is a bounded approximation of $\mathcal{U}_{\text{MI}}$ and further show that it leads to a better approximation of $\mathcal{U}_{\text{MI}}$ in practical scenarios.

PROPOSITION 3.3 ($\mathcal{U}_{\text{MI}}^{\text{TVD}}$ IS BOUNDED BY $\mathcal{U}_{\text{MI}}$). *Let D be a database such that the schema of D is a superset of $P \cup O$. Given a fairness constraint $P \perp O$, the following holds: $\alpha \cdot \mathcal{U}_{\text{MI}}(P, O) \leq \mathcal{U}_{\text{MI}}^{\text{TVD}}(P \perp O, D) \leq \mathcal{U}_{\text{MI}}(P, O)$ where $X = Pr_D(P, O)$, $Y = Pr_D(P)Pr_D(O)$, and $\alpha = \min_{z \in Z, Y(z) > 0} Y(z)$.*

We defer all proofs to the full version of the paper [95].

Though the lower bound could be loose when α is small given a skewed dataset, in practice $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ approximates $\mathcal{U}_{\text{MI}}$ tightly. In the full version of the paper [95] we show the values of $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ compared to $\mathcal{U}_{\text{MI}}$, $\mathcal{U}_{\text{MI}}^{\text{Bayes}}$, and its offset version. The trend of $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ values emulates the trend of $\mathcal{U}_{\text{MI}}$ values better than the other measures. Furthermore, Figure 5b in Section 5.2 shows a similar faithfulness trend of $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ to $\mathcal{U}_{\text{MI}}$ on a synthetic dataset with increasing unfairness.

We now show that $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ satisfies the desired properties. The first two properties can be shown directly, while computability under DP will be proven in a two-part fashion: (1) bounding the sensitivity and range of the measure in the following proposition and (2) providing an ϵ -DP algorithm for computing it in Section 4.

PROPOSITION 3.4 ($\mathcal{U}_{\text{MI}}^{\text{TVD}}$ SATISFIES THE DESIRED PROPERTIES).

- (1) $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ satisfies the Positivity property.
- (2) $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ satisfies the Monotonicity property.
- (3) The range of $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ is $[0, 2|\mathcal{F}|]$ for a set of criteria $\mathcal{F}$.
- (4) The sensitivity of $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ is $\frac{16|\mathcal{F}|}{n}$ for a set of criteria $\mathcal{F}$ and a database of size n .

3.2 Unfairness as Data Repair Cost

Inspired by the field of data repair [3, 10, 18, 39, 68, 83] and previous work on inconsistency measures for integrity constraints [69, 79], including a recent work that allows for their DP computation [74], we use a similar idea to define another unfairness measure.

Definition 3.5 (Data Repair Unfairness). *Given a database D and a fairness criterion of the form $F = P \perp O \mid A$ where A can be empty, the repair unfairness measure is defined as $\mathcal{U}_{\text{R}}(F, D) = |D \dot{-} D_{\text{R}}|$, where D_{R} is the database with the smallest number of removed and added tuples to D that satisfies F and $\dot{-}$ is the symmetric difference.*

It follows that $\mathcal{U}_{\text{R}}$ satisfies the Positivity and Monotonicity properties, its range is $[0, n|\mathcal{F}|]$, and its sensitivity is $|\mathcal{F}|$. However, the problem of finding a minimum data repair is known to be a particularly challenging one, as we shall next elaborate.

Computational challenge. The problem of data repair has been thoroughly studied with different intervention models, such as tuples deletions [41, 67], value update [18, 39, 81], and combinations of tuple additions and deletions [83]. Its computational hardness is well established, even under tuple deletion alone [67]. Although our goal is to measure the size of the repair with deletions and additions and not to find the repair itself, the two problems are computationally equivalent. To address this challenge, in Section 4, we adapt an algorithm from previous work [83] that reduces the problem of computing $\mathcal{U}_{\text{R}}$ to the Max-3SAT problem, and then uses a SAT solver to obtain a repair. Therefore, for $\mathcal{U}_{\text{R}}$ we resort to a proxy measure called $\mathcal{U}_{\text{R}}^{\text{SAT}}$ that is based on prior work [83], not for privacy purposes, but for computational purposes.

Review of the approach from [83]. To overcome intractability, Salimi et al. [83] reduce the computation of an optimal repair satisfying $P \perp O \mid A$ to weighted MaxSAT (which can then be solved by a SAT solver), where $\mathcal{A} = \{P, O, A\}$.²

Define the self-join database $D_{sj} = \Pi_{P,A}(D) \bowtie \Pi_{O,A}(D)$. Then, any minimal repair D' of D satisfies $D' \subseteq D_{sj}$. The reduction converts the tuples in D_{sj} to a CNF formula by adding a clause x_t for every $t \in D$ and a clause $\neg x_t$ for every $t \in D_{sj} \setminus D$. These are the ‘soft clauses’ of the formula. It also adds the clauses $\neg x_{t_1} \vee \neg x_{t_2} \vee x_{t_3}$ for tuples of the form $t_1[p_1, y_1, a]$, $t_2[p_2, y_2, a]$ and $t_3[p_1, y_2, a]$. Intuitively, t_1 and t_2 are the lineage of t_3 . Thus, these are the ‘hard clauses’ of the formula, i.e., they must be satisfied by the obtained assignment. Salimi et al. [83] uses a SAT solver to find an assignment that maximizes the additive weight of the satisfied soft clauses of the constructed CNF formula. The assignment directly corresponds to the tuples that have to be removed and added to satisfy the fairness criterion. Next, we show how we utilize and extend this approach to define $\mathcal{U}_{\text{R}}^{\text{SAT}}$, which is our proxy for $\mathcal{U}_{\text{R}}$.

3.2.1 Defining $\mathcal{U}_{\text{R}}^{\text{SAT}}$. Following previous work [83], we define $\mathcal{U}_{\text{R}}^{\text{SAT}}$ in terms of an assignment to the relevant CNF formula. We begin by defining this CNF formula that D_{sj} is mapped to.

²In practice, the database schema may contain additional attributes that do not appear in the fairness criterion. We therefore compute the repair on the projection to the criterion’s attributes and, when we lift this repaired projection back to a full repair of the original schema, we keep all other attributes as in the original database. Since these additional attributes are unchanged, all propositions that we prove in the sequel with the reduced schema assumption carry over to the full schema.

Definition 3.6 (CNF formula for D_{sj} [83]). *Given a database D whose schema contains $P \cup O \cup A$, and a fairness criterion $F = P \perp O \mid A$, the self-join database CNF formula is defined as*

$$\varphi(D, D_{sj}) = \mathcal{H}(D_{sj}) \wedge \bigwedge_{t \in D} x_t \wedge \bigwedge_{t \in D_{sj} \setminus D} (\neg x_t),$$

where each x_t represents a soft clause for a tuple t and $\mathcal{H}(D_{sj})$ is the set of hard clauses constructed as follows. Let $C(P_1, O_1, P_2, O_2, A) := D_{sj}(P_1, O_1, A) \wedge D_{sj}(P_2, O_2, A)$. That is, C is the set of all tuples formed by joining D_{sj} with itself on the attribute A , pairing tuples with the same A -values and extracting the relevant attributes. For each tuple $t \in C$, a clause of the form $(\neg x_{t_1} \vee \neg x_{t_2} \vee x_{t_3})$ is added to $\mathcal{H}(D_{sj})$, where $t_1 = (P_1, O_1, A)$, $t_2 = (P_2, O_2, A)$, $t_3 = (P_1, O_2, A)$.

We treat the soft clauses $\bigwedge_{t \in D} x_t \wedge \bigwedge_{t \in D_{sj} \setminus D} (\neg x_t)$, as a bag, meaning that soft clauses for two identical tuples (except their IDs) will both appear in $\varphi(D, D_{sj})$.

A feasible assignment for a CNF formula $\varphi(D, D_{sj})$ is an assignment, α , that satisfies all the hard clauses $\mathcal{H}(D_{sj})$.

We now define the repair cost of a database based on an assignment from Definition 3.6. We will then link this notion with the notion of data repair by means of tuple deletions and additions.

Definition 3.7 (Repair cost under assignment). *Given a database D , a fairness criterion F , a CNF formula $\varphi(D, D_{sj})$ and α a feasible assignment for φ , let D_R denote the set of tuples $t \in D_{sj}$ such that x_t is assigned True under the assignment α . We define the cost of repairing D under the assignment α as $\text{dist}(\varphi(D, D_{sj}), \alpha) = |D \div D_R|$. That is, the size of the set of tuples in D whose corresponding variables are assigned to False, or are not present in D and assigned to True.*

The following lemma discusses the direct translation between the number of soft clauses satisfied by an assignment to the CNF formula and the number of changes in the database required for it to satisfy the fairness criterion. In particular, a larger number of satisfied clauses means a smaller change in the original database.

Lemma 3.8. *Given a database D and a fairness criterion, let D_{sj} be the self-join database. Let $\varphi(D, D_{sj})$ be a CNF formula defined according to Definition 3.6 and let α_1 and α_2 be two feasible assignments for φ . Finally, let $\text{dist}(\varphi(D, D_{sj}), \alpha_1)$ and $\text{dist}(\varphi(D, D_{sj}), \alpha_2)$ be defined according to Definition 3.7. If α_1 satisfies more soft clauses than α_2 , then: $\text{dist}(\varphi(D, D_{sj}), \alpha_1) < \text{dist}(\varphi(D, D_{sj}), \alpha_2)$.*

By applying Lemma 3.8, we can define a proxy for $\mathcal{U}_R$, $\mathcal{U}_R^{S\mathcal{A}\mathcal{T}}$, as the repair with the minimum cost based on Definition 3.7.

Definition 3.9 ($\mathcal{U}_R^{S\mathcal{A}\mathcal{T}}$ as a proxy for $\mathcal{U}_R$). *Given a database D , a fairness criterion F , and the CNF formula $\varphi(D, D_{sj})$, the cost of an optimal repair of D through $\varphi(D, D_{sj})$ is*

$$\mathcal{U}_R^{S\mathcal{A}\mathcal{T}}(F, D) := \min_{\alpha \models \mathcal{H}(D_{sj})} \text{dist}(\varphi(D, D_{sj}), \alpha)$$

We detail the properties of $\mathcal{U}_R^{S\mathcal{A}\mathcal{T}}$, showing that it satisfies the required desiderata and its bounded sensitivity relative to its range.

PROPOSITION 3.10 ($\mathcal{U}_R^{S\mathcal{A}\mathcal{T}}$ SATISFIES THE DESIRED PROPERTIES).

- (1) $\mathcal{U}_R^{S\mathcal{A}\mathcal{T}}$ satisfies the Positivity property.
- (2) $\mathcal{U}_R^{S\mathcal{A}\mathcal{T}}$ satisfies the Monotonicity property.
- (3) The range of $\mathcal{U}_R^{S\mathcal{A}\mathcal{T}}$ is $[0, n|\mathcal{F}|]$ for a set of criteria $\mathcal{F}$.
- (4) The sensitivity of $\mathcal{U}_R^{S\mathcal{A}\mathcal{T}}$ is $2|\mathcal{F}|$ for a set of criteria $\mathcal{F}$.

3.3 Unfairness as Top Contributions

Previous work has measured tuple contributions as both measuring inconsistency [69] and as means of explaining results [23, 65, 66, 73]. We adapt this notion to define an unfairness measure that measures individual tuple contribution to unfairness. First, we define the notion of marginal difference for a fairness criterion $F = P \perp O \mid A$ where A can be absent, as follows:

$$\begin{aligned} \mathcal{MD}(F, D, t) = & Pr_D(A = t[A]) \left| Pr_D(P = t[P], O = t[O] \mid A = t[A]) \right. \\ & \left. - Pr_D(P = t[P] \mid A = t[A]) Pr_D(O = t[O] \mid A = t[A]) \right|, \end{aligned}$$

where $Pr_D(A = t[A]) = 1$ when A is absent.

We now define the $\mathcal{U}_{TC}$ measure using the $\mathcal{MD}$ notion.

Definition 3.11 (Top- k Tuple Contribution Unfairness). *Given a database D , a natural number k , and a fairness criterion of the form $F = P \perp O \mid A$, define top- k as the set of k tuples with the largest $\mathcal{MD}$ values. We assume that each tuple in the database has a unique ID, so that tuples with the same values but different IDs can both appear in the top- k . Then, the tuple contribution unfairness measure is defined as $\mathcal{U}_{TC}(F, D) = \sum_{t \in \text{top-}k} \mathcal{MD}(F, D, t)$.*

While Definition 3.11 resembles Definition 3.2 as a cumulative sum of residual tuple differences, they diverge when databases have similar sums but different contribution distributions, e.g., a ‘long tail’ of outliers. We demonstrate this distinction empirically in [95].

In the following proposition, we assume that if $P \perp O \mid A \in \mathcal{F}$, all values of A occur more than once in the dataset. This was the case in all real-world datasets included in our experiments.

PROPOSITION 3.12 ($\mathcal{U}_{TC}$ SATISFIES THE DESIRED PROPERTIES). *Given a natural number k , the following holds for $\mathcal{U}_{TC}$:*

- (1) $\mathcal{U}_{TC}$ satisfies the Positivity property.
- (2) $\mathcal{U}_{TC}$ satisfies the Monotonicity property.
- (3) The range of $\mathcal{U}_{TC}$ is $[0, \min\{\frac{k}{4}, 2\}|\mathcal{F}|]$ for a set of criteria $\mathcal{F}$.
- (4) The sensitivity of $\mathcal{U}_{TC}$ is $\frac{3k}{n}|\mathcal{F}|$ for a set of unconditional $\mathcal{F}$, and $\frac{2k}{n}|\mathcal{F}|$ for a set of conditional $\mathcal{F}$.

Practical use of the measures. $\mathcal{U}_{MI}^{TVD}$, $\mathcal{U}_R^{S\mathcal{A}\mathcal{T}}$, and $\mathcal{U}_{TC}$ satisfy monotonicity with a lower bound of 0 (absolute fairness) and an upper bound (Propositions 3.4, 3.10 and 3.12), enabling detection of both fair and extremely unfair cases, demonstrated by one of our use cases in Section 5.2. For nuanced assessment, users can calibrate against a likely-satisfied baseline criterion; e.g., in the Adult dataset, using $\text{sex} \perp \text{race}$ as a reference point.

Relation to other suggested properties. Previous work [69] has also proposed the *continuity* and *progression* properties for inconsistency measures. Underlying both properties is a dynamic data repair process with an associated operation set O , e.g., a single tuple deletion and/or modification, and a parameter $\delta \geq 1$.

Informally, continuity limits the rate of inconsistency change caused by a single operation from O . In our setting, for every two databases D_1, D_2 , criteria $\mathcal{F}$, and operation $o_1 \in O$, there exists $o_2 \in O$ such that $|\mathcal{U}(\mathcal{F}, D_1) - \mathcal{U}(\mathcal{F}, o_1(D_1))| \leq \delta |\mathcal{U}(\mathcal{F}, D_2) - \mathcal{U}(\mathcal{F}, o_2(D_2))|$. Assuming O allows a single tuple modification or deletion, we can derive a similar property from the sensitivity analysis of the measures (Table 1), which guarantees that the rate of change is bounded

for neighboring datasets, where D and $D' = o_1(D)$ are neighbors: $|\mathcal{U}(\mathcal{F}, D_1) - \mathcal{U}(\mathcal{F}, o_1(D_1))| \leq \Delta_{\mathcal{U}}$.

Progression states that in any case where a database violates the fairness criteria, the repair process always allows for some path towards more database consistency, i.e., there is always an operation $o \in O$ such that inconsistency is reduced after applying o . As opposed to the integrity constraints considered in [69] which are anti-monotonic, fairness criteria do not have this property. Therefore, if O only allows for tuple deletions for example, progression does not necessarily hold. To see this, consider the criterion $\text{sex} \perp \text{income} > 50K$ and a database where all individuals have $\text{sex} = M$ and $\text{income} > 50K = 1$. This database violates the fairness criterion but any tuple deletion will not reduce the unfairness expressed via $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ and $\mathcal{U}_{\text{TC}}$. For $\mathcal{U}_{\text{R}}^{\text{SAT}}$, consider the criteria $\mathcal{F} = \{A \perp B, C \perp D\}$ and the database D with the schema (A, B, C, D) and tuples $(1, 1, a, x), (1, 0, b, y), (0, 1, b, y), (0, 0, a, x)$. D satisfies $A \perp B$ but violates $C \perp D$. Any single tuple deletion will make the database violate $A \perp B$, necessitating at least a single increment of the $\mathcal{U}_{\text{R}}^{\text{SAT}}$ measure.

4 PRIVATE MEASURE COMPUTATION

We describe the algorithms that compute the unfairness measures and comply with DP, showing the third desideratum from Section 3.

4.1 Computing $\mathcal{U}_{\text{MI}}^{\text{TVD}}$

The pseudocode for computing $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ is summarized in Algorithm 1. In line 1, the algorithm initializes to zero the variable that will accumulate $\mathcal{U}_{\text{MI}}^{\text{TVD}}(F, D)$ for all F in the set of criteria $\mathcal{F}$. For each fairness criterion of the form $F = P \perp O \mid A$ in the set of criteria $\mathcal{F}$, in line 3 the algorithm computes empirical probabilities derived from the dataset D : the conditional joint probability $Pr_D(P, O \mid A)$ and the conditional marginal probabilities $Pr_D(P \mid A)$ and $Pr_D(O \mid A)$. In case A is not given, we assume that $Pr_D(P, O \mid A)$ is equal to $Pr_D(P, O)$, $Pr_D(P \mid A)$ is equal to $Pr_D(P)$, and $Pr_D(O \mid A)$ is equal to $Pr_D(O)$.

The algorithm splits into two cases: unconditional (no A), computing TVD directly in line 5; and conditional, computing $Pr_D(A = a)$ for each a in line 7, then summing the probability-weighted conditional TVDs in line 9. Finally, the algorithm computes $\mathcal{U}_{\text{MI}}^{\text{TVD}}(F, D)$ as $2 \cdot \text{TVD}^2$ (Definition 3.2) and updates the cumulative sum with this value in line 10. After computing $\mathcal{U}_{\text{MI}}^{\text{TVD}}$, the algorithm applies the Laplace mechanism in line 11 by adding noise according to the sensitivity $\frac{16|\mathcal{F}|}{n}$ (Item 4 in Proposition 3.4), and the privacy budget ϵ . The resulting value $\widehat{\mathcal{U}_{\text{MI}}^{\text{TVD}}}(\mathcal{F}, D)$ is then returned.

The complexity of Algorithm 1 is $O(|\mathcal{F}|n)$. We defer the detailed complexity analysis to the full version of the paper [95].

We next show that Algorithm 1 satisfies DP with bounded error, and thus $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ satisfies the third property in Section 3.

PROPOSITION 4.1 (DP AND ERROR BOUND OF ALGORITHM 1).

- (1) Algorithm 1 is ϵ -DP.
- (2) For a database D and a set of fairness criteria $\mathcal{F}$, Algorithm 1 returns $\widehat{\mathcal{U}_{\text{MI}}^{\text{TVD}}}(\mathcal{F}, D)$ such that for any $\epsilon > 0$, it holds that

$$\mathbb{E} \left[\left| \widehat{\mathcal{U}_{\text{MI}}^{\text{TVD}}}(\mathcal{F}, D) - \mathcal{U}_{\text{MI}}^{\text{TVD}}(\mathcal{F}, D) \right| \right] = \frac{16|\mathcal{F}|}{n\epsilon}$$

Algorithm 1: Compute $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ under DP

Input: Database D ; set of fairness criteria $\mathcal{F}$; privacy budget ϵ

Output: $\widehat{\mathcal{U}_{\text{MI}}^{\text{TVD}}}(\mathcal{F}, D)$

```

1  $\mathcal{U}_{\text{MI}}^{\text{TVD}}(\mathcal{F}, D) \leftarrow 0$ ;
2 foreach  $F = (P \perp O \mid A) \in \mathcal{F}$  do
3   Compute  $Pr_D(P = p, O = y \mid A = a)$ ,
    $Pr_D(P = p \mid A = a)$ ,  $Pr_D(O = y \mid A = a)$ ;
   /* Unconditional criterion */
4   if  $A$  is  $\emptyset$  then
5      $\text{TVD} \leftarrow \frac{1}{2} \sum_{p \in P, y \in O} |Pr_D(P = p, O = y) - Pr_D(P = p) Pr_D(O = y)|$ ;
   /* Conditional criterion */
6   else
7     Compute  $Pr_D(A = a)$  for every  $a \in A$ ;
8      $\text{TVD} \leftarrow \sum_{a \in A} Pr_D(A = a) \cdot \left( \frac{1}{2} \sum_{p \in P, y \in O} |Pr_D(P = p, O = y \mid A = a) - Pr_D(P = p \mid A = a) Pr_D(O = y \mid A = a)| \right)$ ;
9    $\mathcal{U}_{\text{MI}}^{\text{TVD}}(\mathcal{F}, D) \leftarrow \mathcal{U}_{\text{MI}}^{\text{TVD}}(\mathcal{F}, D) + 2 \cdot \text{TVD}^2$ ;
11 return  $\mathcal{U}_{\text{MI}}^{\text{TVD}}(\mathcal{F}, D) + \text{Lap}\left(0, \frac{16|\mathcal{F}|}{n\epsilon}\right)$ ;

```

4.2 Computing $\mathcal{U}_{\text{R}}^{\text{SAT}}$

Algorithm 2 combines the weighted CNF conversion from [83] with a SAT solver to solve the weighted MaxSAT problem, using the noise scale determined in Section 3.2.1 to ensure DP.

In line 1, the algorithm initializes the variable that will accumulate $\mathcal{U}_{\text{R}}^{\text{SAT}}(F, D)$ for all F in the set $\mathcal{F}$. For each fairness criterion $F = P \perp O \mid A$ in the set $\mathcal{F}$, the code block 3 runs the algorithm from [83]. This block first initializes φ to an empty set, and then adds soft and hard clauses as in definition 3.6. Then, the algorithm calls a SAT solver in line 4 to obtain an assignment α that satisfies all the hard clauses and maximizes the number of satisfied soft clauses in φ . From this assignment, the algorithm constructs the repaired database in line 5, which has all the tuples that α assigns True. In line 5 it computes $\mathcal{U}_{\text{R}}^{\text{SAT}}(F, D)$ according to Definition 3.9, and updates the cumulative sum with this value.

Finally, the algorithm applies the Laplace mechanism in line 6 and adds noise according to the sensitivity $2|\mathcal{F}|$ (see Item 4 in Proposition 3.10). The noisy version of $\mathcal{U}_{\text{R}}^{\text{SAT}}$ is then returned.

The complexity of Algorithm 2 is $O(|\mathcal{F}|(n^4 + \text{SAT}))$. We defer the detailed complexity analysis to the full version [95].

PROPOSITION 4.2 (DP AND ERROR BOUND ON ALGORITHM 2).

- (1) Algorithm 2 is ϵ -DP.
- (2) For a database D and a set of fairness criteria $\mathcal{F}$, Algorithm 2 returns $\widehat{\mathcal{U}_{\text{R}}^{\text{SAT}}}(\mathcal{F}, D)$ such that for any $\epsilon > 0$, it holds that $\mathbb{E} \left[\left| \widehat{\mathcal{U}_{\text{R}}^{\text{SAT}}}(\mathcal{F}, D) - \mathcal{U}_{\text{R}}^{\text{SAT}}(\mathcal{F}, D) \right| \right] = \frac{2|\mathcal{F}|}{\epsilon}$.

4.3 Computing $\mathcal{U}_{\text{TC}}$

We describe the pseudocode in Algorithm 3 for computing $\mathcal{U}_{\text{TC}}$. In line 1, the algorithm initializes to zero the variable that will accumulate $\mathcal{U}_{\text{TC}}(F, D)$ for all F in the set of criteria $\mathcal{F}$. For each

Algorithm 2: Compute $\mathcal{U}_R^{S\mathcal{AT}}$ under DP

Input: Database D ; set of fairness criteria $\mathcal{F}$; privacy budget ε

Output: $\mathcal{U}_R^{S\mathcal{AT}}(\mathcal{F}, D)$

```

1  $\mathcal{U}_R^{S\mathcal{AT}}(\mathcal{F}, D) \leftarrow 0$ ;
2 foreach  $F = (P \perp O \mid A) \in \mathcal{F}$  do
    /* Algorithm from [83] */
     $D_{sj}(P_1, O_2, A) \leftarrow D(P_1, O_1, A) \bowtie D(P_2, O_2, A)$ ;
     $\varphi \leftarrow \emptyset$ ;
    foreach  $t \in D_{sj}$  do
        if  $t \in D$  then
            Add the soft clause  $x_t$  to  $\varphi$ ;
        if  $t \notin D$  then
            Add the soft clause  $\neg x_t$  to  $\varphi$ ;
     $C(P_1, O_1, P_2, O_2, A) \leftarrow D_{sj}(P_1, O_1, A) \wedge D_{sj}(P_2, O_2, A)$ ;
    foreach  $t \in C$  do
         $t_1 \leftarrow t(P_1, O_1, A), t_2 \leftarrow t(P_2, O_2, A), t_3 \leftarrow t(P_1, O_2, A)$ ;
        Add the hard clause  $(\neg x_{t_1} \vee \neg x_{t_2} \vee x_{t_3})$  to  $\varphi$ ;
3  $\alpha \leftarrow \text{Solver}(\varphi)$ ;
4  $D_R \leftarrow \{t \mid \alpha(x_t) = \text{True}\}$ ;
5  $\mathcal{U}_R^{S\mathcal{AT}}(\mathcal{F}, D) \leftarrow \mathcal{U}_R^{S\mathcal{AT}}(\mathcal{F}, D) + |D \div D_R|$ ;
6 return  $\mathcal{U}_R^{S\mathcal{AT}}(\mathcal{F}, D) + \text{Lap}\left(0, \frac{2|\mathcal{F}|}{\varepsilon}\right)$ ;

```

fairness criterion of the form $F = P \perp O \mid A$ in the set of criteria $\mathcal{F}$, in line 3 the algorithm computes empirical probabilities derived from the dataset D : the conditional joint probability $Pr_D(P, O \mid A)$ and the conditional marginal probabilities $Pr_D(P \mid A)$ and $Pr_D(O \mid A)$. In case A is not given, we assume that $Pr_D(P, O \mid A)$ is equal to $Pr_D(P, O)$, $Pr_D(P \mid A)$ is equal to $Pr_D(P)$, and $Pr_D(O \mid A)$ is equal to $Pr_D(O)$. Then, the algorithm iterates over every tuple t in D and in line 5 it computes the marginal difference $\mathcal{MD}(F, D, t)$ (defined in Section 3.3). The resulting value quantifies the contribution of each tuple t to the deviation of the observed joint probability from the joint probability in the case when the independence defined by F would hold. After computing marginal differences for all tuples in the dataset, the algorithm sorts them in descending order in line 6. In line 7 it adds the sum of the k largest marginal differences to the global value $\mathcal{U}_{TC}(\mathcal{F}, D)$. Intuitively, for each criterion in $\mathcal{F}$, it adds the violation influence of the k most influential tuples to the total score. The algorithm performs all the previous stages for every F in $\mathcal{F}$. After iterating over all of them, the algorithm adds Laplace noise in line 12 according to the sensitivity $\frac{7k}{n}|\mathcal{F}|$ if there is a conditional $F \in \mathcal{F}$, and $\frac{3k}{n}|\mathcal{F}|$ otherwise (Item 4 in Proposition 3.12). This noisy value is returned as the DP version of $\mathcal{U}_{TC}$.

The complexity of Algorithm 3 is $O(|\mathcal{F}|n \log n)$. We defer the detailed complexity analysis to the full version [95].

PROPOSITION 4.3 (DP AND ERROR BOUND OF ALGORITHM 3).

- (1) Algorithm 3 is ε -DP.
- (2) For a database D , Algorithm 3 returns $\widetilde{\mathcal{U}}_{TC}(\mathcal{F}, D)$ such that for any $\varepsilon > 0$, it holds that

$$\mathbb{E} \left[\left| \widetilde{\mathcal{U}}_{TC}(\mathcal{F}, D) - \mathcal{U}_{TC}(\mathcal{F}, D) \right| \right] = \frac{3k|\mathcal{F}|}{n\varepsilon}$$

Algorithm 3: Compute $\mathcal{U}_{TC}$ under DP

Input: Database D ; set of fairness criteria $\mathcal{F}$; accuracy parameter $k \in \mathbb{N}$; privacy budget ε

Output: $\widetilde{\mathcal{U}}_{TC}(\mathcal{F}, D)$

```

1  $\mathcal{U}_{TC}(\mathcal{F}, D) \leftarrow 0$ ;
2 foreach  $F = (P \perp O \mid A) \in \mathcal{F}$  do
3   Compute  $Pr_D[P = p, O = y \mid A = a]$ ,
    $Pr_D[P = p \mid A = a]$ ,  $Pr_D[O = y \mid A = a]$ ;
4   foreach  $t \in D$  do
5      $\mathcal{MD}(F, D, t) \leftarrow Pr_D[A = a] \left| Pr_D[P = p, O = y \mid A = a] - Pr_D[P = p \mid A = a] Pr_D[O = y \mid A = a] \right|$ ;
6   Sort  $\mathcal{MD}(F, D, t)$  for all  $t \in D$  in descending order;
7    $\mathcal{U}_{TC}(\mathcal{F}, D) \leftarrow \mathcal{U}_{TC}(\mathcal{F}, D) + \sum_{t \in \text{top-}k} \mathcal{MD}(F, D, t)$ ;
8 if  $\exists F = (P, O \mid A) \in \mathcal{F}$  such that  $A$  is given then
9    $\Delta \leftarrow \frac{7k}{n}$ ;
10 else
11    $\Delta \leftarrow \frac{3k}{n}$ ;
12 return  $\mathcal{U}_{TC}(\mathcal{F}, D) + \text{Lap}\left(0, \frac{|\mathcal{F}| \cdot \Delta}{\varepsilon}\right)$ ;

```

for a set of unconditional fairness criteria $\mathcal{F}$, and

$$\mathbb{E} \left[\left| \widetilde{\mathcal{U}}_{TC}(\mathcal{F}, D) - \mathcal{U}_{TC}(\mathcal{F}, D) \right| \right] = \frac{7k|\mathcal{F}|}{n\varepsilon}$$

for a set of conditional fairness criteria.

5 EXPERIMENTS

We have experimentally studied our measures and the algorithms computing them by answering the following questions: (1) What is the ability of the measures to detect different levels of unfairness? (2) How do our algorithms scale? (3) What is the effect of the privacy budget on measure accuracy? (4) How do the measures perform in terms of faithfulness and under parameter variation?

Summary of our findings. Our experiments confirm that the proposed unfairness measures are practical for real-world fairness assessment, serving both as pre-query trust indicators and as alternatives to model-based estimation. The measures detect unfairness correctly and proportionately (Figure 5a), all three increase monotonically with the Demographic Parity gap, with $\mathcal{U}_R^{S\mathcal{AT}}$ increasing near-linearly and $\mathcal{U}_{MI}^{TVD}$, $\mathcal{U}_{TC}$ increasing approximately logarithmically. Their relative ordering agrees with the observed query output disparities (Section 5.2), and $\mathcal{U}_{MI}^{TVD}$ closely tracks $\mathcal{U}_{MI}$ (Figure 5b). All algorithms scale steadily with tuple count and fairness criteria (Figure 3), where Algorithm 3 is the fastest, followed by Algorithm 1, both completing in seconds on large datasets, while Algorithm 2 is $10^2 - 10^3$ times slower due to its weighted MaxSAT solver, yet remains valuable for capturing the most nuanced unfairness notion.

Additional experiments detailed in [95] show that (i) $\mathcal{U}_{MI}^{TVD}$ is a good proxy for $\mathcal{U}_{MI}$, (ii) $\mathcal{U}_{TC}$ increases monotonically with k up to some value, indicating that a dominant subset of tuples drives unfairness, and (iii) our heuristic for $\mathcal{U}_R^{S\mathcal{AT}}$ improves runtime by over 50% while preserving relative measure values.

5.1 Experimental Setup

All algorithms and experiments were implemented in Python using standard packages such as networkx, numpy, pandas, and z3-solver. The experiments ran on a machine with Intel Core i5-12400F. For each data point in every experiment, we performed 10 independent repetitions, unless stated otherwise, and averaged the result. **If one of these repetitions took longer than 24 hours to complete, we did not include this data point in the graph.** For experiments with fixed data size, we randomly sampled up to 100K tuples from each dataset, or used the entire dataset if it is smaller, ensuring that the assumption of multiplicity in the A values holds (Proposition 3.12).

Datasets. We used the following five datasets in our experiments.

- **Adult:** A dataset downloaded from ³, containing 15 attributes and 48,843 tuples. It contains personal data about individuals, including their sex, race, marital status, whether their income is over 50K, etc. Some attributes were renamed for clarity (e.g., the binary attribute income was renamed to income>50K).
- **IPUMS-CPS:** A population survey dataset published by the U.S. Census Bureau [34] and downloaded from ⁴, containing 8 attributes and 1,048,576 tuples, with data from 2011 till 2019. The dataset, like the Adult dataset, contains personal data about individuals.
- **Stackoverflow** ⁵: A dataset containing 114 attributes and 65,438 tuples. The dataset contains the results of the Stackoverflow developers survey for 2024, and includes languages they use, where they learned to code, information on the companies they work at, etc.
- **Compas** ⁶: A dataset containing 47 attributes and 11,757 tuples. It contains the criminal history, jail and prison time, demographics and Compas risk scores for defendants from Broward County from 2013 and 2014. The data is usually used to predict whether a person will re-offend within the next 2 years.
- **Healthcare** ⁷: A dataset containing 11 attributes and 1000 tuples. It contains personal data of patients, such as the income, number of complications, county, race, age.

All datasets were preprocessed so that negative numerical values were replaced by zeros since their domains are nonnegative, categorical values were encoded, **with missing or undefined values being treated as a separate category**. For the IPUMS-CPS dataset, the attribute AGE was discretized as 10 years per range, e.g., [0, 10] is considered a single value. We only included tuples with attribute INCTOT (total income) smaller than 200K to avoid outliers, as in [88].

Fairness criteria. In Section 2.2, we formalized fairness criteria as conditional independence statements, instantiated per dataset by choosing concrete protected, response, and admissible attributes; the resulting criteria are listed in Table 2 and numbered 1–4 in the figures. The four criteria reflect natural desiderata for protected attributes (e.g., sex or race) to be independent from the predicted attribute (e.g., income).

Algorithm variations and optimizations. Since there are no existing baselines for unfairness measures, we compared Algorithms 1

Table 2: Fairness criteria per dataset.

Dataset	Fairness criteria
Adult	(1) education-num $\perp$ income>50K (2) sex $\perp$ income>50K (3) race $\perp$ income>50K (4) sex $\perp$ income>50K hours-per-week
IPUMS-CPS	(1) HEALTH $\perp$ INCTOT EDUC (2) HEALTH $\perp$ OCC EDUC (3) HEALTH $\perp$ MARST AGE (4) HEALTH $\perp$ INCTOT AGE
Stackoverflow	(1) Country $\perp$ RemoteWork Employment (2) Age $\perp$ PurchaseInfluence OrgSize (3) Country $\perp$ MainBranch YearsCodePro (4) Age $\perp$ MainBranch EdLevel
Compas	(1) race $\perp$ is_recid age_cat (2) sex $\perp$ is_recid priors_count (3) race $\perp$ decile_score c_charge_degree (4) sex $\perp$ v_decile_score age_cat
Healthcare	(1) race $\perp$ complications age_group (2) smoker $\perp$ complications age_group (3) race $\perp$ income county (4) smoker $\perp$ income num_children

to 3 with their several variations and employed a heuristic for Algorithm 2:

- **Non-private versions:** In Figure 4, we compared Algorithms 1 to 3 with varying privacy budgets from 0.1 to 10 with their non-private counterparts, i.e., setting $\epsilon = \infty$. For the rest of the experiments, we use a privacy budget of $\epsilon = 1$, unless otherwise specified.
- **k in Algorithm 3:** We fixed $k = 500$ in all experiments.
- **Heuristic for Algorithm 2:** Due to scalability issues of the SAT solver in line 4 of Algorithm 2, we employed a heuristic that partitions the database to consecutive chunks of 100 tuples, then runs Algorithm 2 separately for each chunk and each criterion, and sums up the results to yield an estimate of the $\mathcal{U}_R^{\text{SAT}}$ value. Assuming that the tuple order is preserved in neighboring databases, as is standard for bounded DP [77], the sensitivity of this heuristic remains identical to $\Delta_{\mathcal{U}_R^{\text{SAT}}}$, as proven in [95].

Relative L1 error. In some of the experiments we measured relative L1 error of the algorithms. For a given ϵ , we define the relative error as $\frac{|X - Y|}{\max(Y, e^{-100})}$, where X is the output of the algorithm with privacy budget ϵ , and Y is the output of the algorithm without privacy. The use of e^{-100} in the denominator is meant to avoid division by 0.

5.2 Use Cases for the Measures

We next show three case studies for our unfairness measures in the context of (1) data exploration (2) ML models, and (3) increasing data unfairness.

Assessing dataset biases in data exploration. Suppose we are interested in understanding the incomes of different sexes and different races using the Adult dataset with queries q_1, q_2 , that compute average income across groups defined by sex and race, respectively (see [95] for the their SQL description). Beforehand, it is important to know whether the dataset is reliable for drawing conclusions from, and to what extent. To do so, we computed the three unfairness measures with the criteria sex $\perp$ income>50K and race $\perp$ income>50K, corresponding to q_1 and q_2 , respectively. The first two rows in Table 3 show the measures’ results for the above fairness criteria and indicate a more pronounced bias in income

³<https://archive.ics.uci.edu/dataset/2/adult>

⁴<https://cps.ipums.org/cps>

⁵<https://kaggle.com/datasets/berkayalan/stack-overflow-annual-developer-survey-2024>

⁶<https://github.com/propublica/compas-analysis/blob/master/compas-scores.csv>

⁷https://github.com/stefan-grafberger/mlinspect/tree/master/example_pipelines

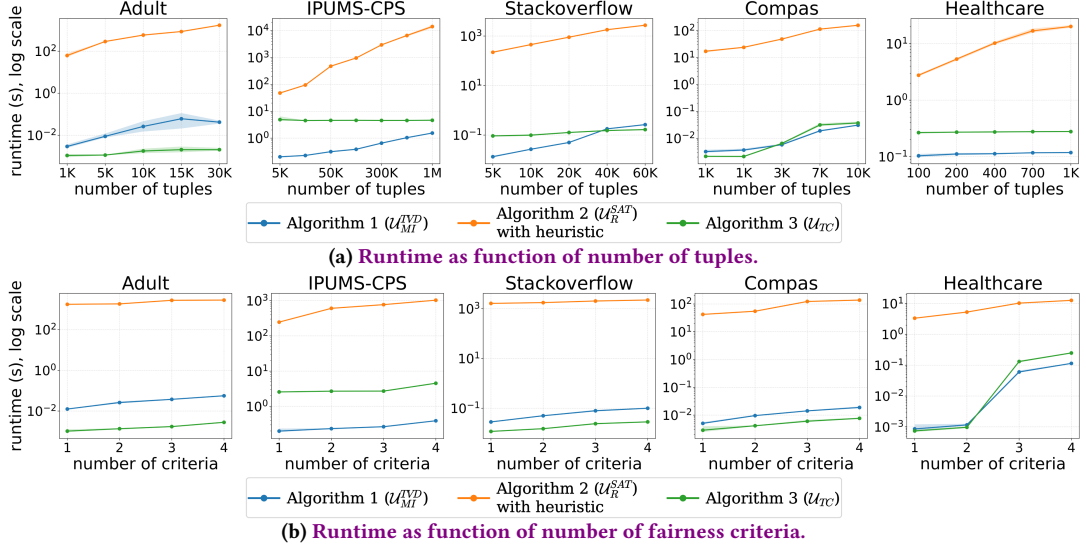


Figure 3: Runtime analysis of the algorithms for the datasets and criteria in Table 2.

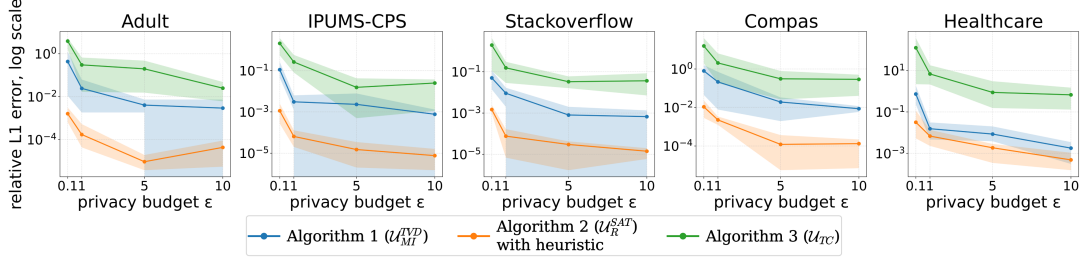


Figure 4: Relative L1 error as function of privacy budget for the datasets and criteria from Table 2.

Table 3: Average unfairness-measure values for the fairness criteria on the Adult and Compas datasets. Darker red indicate higher measure value.

Dataset	Criterion corresponding to (Query)	$\mathcal{U}_{MI}^{TVD}$	$\mathcal{U}_R^{SAT}$	$\mathcal{U}_{TC}$
Adult	sex $\perp$ income>50K (q_1)	0.015	4573.8	0.175
	race $\perp$ income>50K (q_2)	0.002	1227.8	0.053
	race $\perp$ decile_sc age_cat (q_3)	0.035	2517.2	0.263
Compas	age_cat $\perp$ decile_sc race (q_4)	0.049	2704.8	0.310
	race $\perp$ decile_sc c_charge_d. (q_5)	0.039	2662.8	0.282
	c_charge_d. $\perp$ decile_sc race (q_6)	0.015	1897.7	0.169

with respect to sex than race, which may be used as a telltale sign that there is a large income bias based on sex in the dataset. Indeed, we ran the queries and got the results (the results appear in [95]). In q_1 , the proportion of males with income above \$50K (0.304) is nearly three times that of females (0.109), while q_2 shows smaller disparities across race groups (the biggest disparity is between 0.117 for Amer-Indian-Eskimo and 0.269 for Asian-Pac-Islander).

Now consider the analysis of the decile scores (which denote the estimated risk to reoffend, ranging from 1 to 10) using the Compas dataset with queries q_3 – q_6 (see [95] for their SQL description). Specifically, q_3 , given here as an example, computes, for each age group, the average decile score aggregated over race groups:

```
SELECT age_cat, AVG(avg_d) AS avg_d
FROM (SELECT race, age_cat, AVG(decile_score) AS avg_d
```

```
FROM Compas GROUP BY race, age_cat) t
GROUP BY age_cat;
```

Query q_4 reverses the order in q_3 and computes, for each race group, the average decile score aggregated over age groups. q_5 computes, for each race group, the median decile score aggregated over charge degrees, while q_6 computes, for each charge degree, the median decile score aggregated over race groups. Prior to running the queries, we ran the unfairness measures with the criteria in Table 3, corresponding to the queries. The queries map to the criteria as follows. Suppose we have a criterion of the form $S \perp Y | A$, where Y is the attribute of the disparity (decile_score, shortened as decile_sc, for all queries), S is the attribute over which disparities are checked, and A is the attribute being conditioned on, then each query can be mapped to a criterion in Table 3. E.g., for q_3 , it holds that $S = \text{race}$, $A = \text{age_cat}$, $Y = \text{decile_score}$, since the query examines how decile scores vary across race within age groups.

Prior to running the queries, we computed the unfairness measures for the corresponding criteria (Table 3), revealing that age-related disparities in decile scores (conditioned on race) are the most pronounced, followed by race-related disparities conditioned on charge degree, while charge-degree disparities conditioned on race are the least prominent.

The query results, detailed in [95], align with the measure values. q_4 (row 4, highest) shows the largest disparity in decile_score across age_cat groups: 2.970 for unidentified race vs. 5.182 for African-Americans. q_5 (row 5) shows slightly smaller disparities,

with median scores from 1 (unidentified race) to 4 (African-Americans and Native Americans). q_3 (row 3) shows similar but smaller variation across age groups, ranging from 2.636 (older than 45) to 5.201 (younger than 25). q_6 (row 6, lowest) exhibits the least variation, with median scores from 2 (misdemeanor) to 3.5 (felony).

Overall, the measures can alert us to query results that may be biased, when analyzing and drawing conclusions from a dataset.

Table 4: Fairness values (Demographic Parity gaps) of privately-trained variations of Random Forest (RF) and Neural Network (NN) on the Adult dataset.

Model	sex $\perp$ income>50K		race $\perp$ sex	
	Accuracy	DP gap	Accuracy	DP gap
RF, 10 trees	78.91%	0.058	78.22%	0.213
RF, 50 trees	78.24%	0.037	78.50%	0.235
RF, 100 trees	78.41%	0.044	77.54%	0.234
RF, 200 trees	77.91%	0.032	76.06%	0.188
NN, 1 hidden layer	84.16%	0.197	80.37%	0.216
NN, 2 hidden layers	83.82%	0.201	80.03%	0.236
NN, 3 hidden layers	82.40%	0.225	79.33%	0.231

A complementary view to Machine Learning models. To examine how our measures complement ML models in quantifying unfairness, we trained a variety of models with two different classes and hyperparameters to predict income>50K and sex based on the other attributes in the Adult dataset. We measured the Demographic Parity gap of the predictions of income>50K with respect to sex, and the gap in the predictions of sex with respect to race. The latter serves as a baseline, as in real-world settings and intuitively these attributes are expected to be independent, and thus any observed dependence indicates bias. The models, accuracy rates and gaps are detailed in Table 4. We observe that the models consistently estimate a stronger dependence between sex and race than between income>50K and sex. Moreover, despite all models achieving similar accuracy (76.06%-84.16%), the Demographic Parity gaps vary substantially both across model classes (0.032-0.058 for Random Forest, 0.197-0.255 for Neural Network) and within a single class (for Random Forest, 0.032-0.058 for sex and income>50K, and 0.188-0.235 for race and sex), making it difficult to reliably compare the two disparities. Conversely, evaluating the unfairness measures directly on the Adult dataset for the corresponding criteria, sex $\perp$ income>50K and race $\perp$ sex generates the intuitive results consistent with prior work [89, 92] that indicate a strong dependence between sex and income in the Adult dataset: 0.014 and 0.002 for $\mathcal{U}_{MI}^{TVD}$, 4424 and 1432 for $\mathcal{U}_R^{SAT}$, and 0.164 and 0.076 for $\mathcal{U}_{TC}$, respectively.

Finally, the models are limited to assessing unfairness only for the attribute they were trained for. For example, if we compute the Demographic Parity gap for race given sex using a model trained to predict income>50K, we obtain a value of 0.044 for a Random Forest with 200 trees. In contrast, when the same model is trained to predict sex (i.e., aligned with the criterion), the corresponding Demographic Parity gap is 0.188. In contrast, the unfairness measures can run over arbitrary fairness criteria and thus form a complementary alternative to model-based fairness evaluation.

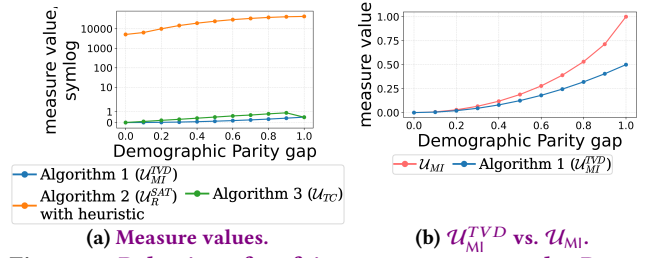


Figure 5: Behavior of unfairness measures as the Demographic Parity gap in the dataset gradually increases.

Detecting varying unfairness. We studied how the proposed unfairness measures react to gradually increasing unfairness in a dataset. We constructed a binary synthetic dataset with attributes sex and income>50K containing 100,000 tuples, with an equal number of male and female records. Initially, the dataset was completely fair (i.e., 50% of each sex group had income>50K = 1). We then progressively flipped an increasing fraction of male records to have income above 50K and female records to have income below 50K, thereby monotonically increasing the demographic parity gap $|Pr_D(\text{income} > 50K \mid \text{sex} = M) - Pr_D(\text{income} > 50K \mid \text{sex} = F)|$. In Figure 5a, we plot the values of the measures as a function of the demographic parity gap. We observe that $\mathcal{U}_R^{SAT}$ (as computed by Algorithm 2 with our heuristic) exhibits almost linear growth as the demographic parity gap increases, while $\mathcal{U}_{MI}^{TVD}$ (as computed by Algorithm 1) and $\mathcal{U}_{TC}$ (as computed by Algorithm 3) exhibit growth that appears logarithmic. $\mathcal{U}_{TC}$ also decreases slightly toward the right-hand side of the graph, since when the dataset becomes fully polarized, $\mathcal{MD}$ for all tuples become similar (i.e., there are no outliers), and so the $top-k$ sum diminishes. Additionally, Figure 5b shows that on the same synthetic dataset $\mathcal{U}_{MI}^{TVD}$ closely follows the trend observed for $\mathcal{U}_{MI}$. Overall, all measures correctly detect small changes in unfairness in the dataset.

5.3 Scalability

Figure 3 depicts the effect of data size and the number of fairness criteria on the runtime of the algorithms. In Figure 3a, we plot the runtime of the algorithms for each dataset as a function of the number of tuples. Across all datasets, the computation of Algorithm 2 is consistently the slowest. The runtime increase of Algorithm 2 stems from its dependence on the weighted MaxSAT solver used to find the minimal repair. However, this step also enables it to capture a more nuanced notion of unfairness than the other measures.

Algorithm 3 is frequently the fastest, with an exception being the IPUMS-CPS dataset, on which the runtime is higher and mostly insensitive to the number of tuples. This is attributed to the fact that, as formulated in its complexity analysis [95], Algorithm 3 uses an empirical contingency table over the attribute sets, whose cardinalities are higher for IPUMS-CPS than for the other datasets. Specifically, it is sometimes faster than Algorithm 1 since the latter has to perform additional manipulations to compute the conditional TVD. Finally, Algorithm 1 is on average faster than Algorithm 2 but slower than Algorithm 3. The runtimes of all the algorithms increase steadily with the number of tuples. Still, runtimes for smaller numbers of tuples are sometimes higher and more volatile because they are more sensitive to randomness in the sampled data.

In Figure 3b, we show the runtime of the algorithms for each dataset as a function of the number of fairness criteria. For each dataset, we begin with only the first criterion in Table 2 and then progressively add more, such that the final measurement includes all criteria for that dataset. We can see that, for all the algorithms, increasing the number of criteria inflates the runtime more gradually than increasing the number of tuples.

Still, the runtime for the Healthcare dataset varies significantly between two and three criteria when computing $\mathcal{U}_{MI}^{TVD}$ and $\mathcal{U}_{TC}$ due to the fact that the third criterion groups tuples by county, which has much higher cardinality than age_group. This increases the size of the empirical distributions computed by Algorithms 1 and 3, leading to higher cost. In contrast, Algorithm 2 is less affected, since its runtime is dominated by the MaxSAT solver.

Overall, Algorithm 3 is the least affected by increasing the number of tuples and the number of criteria for moderately dense datasets, and it is the fastest algorithm in almost all settings. However, its runtime increases noticeably for sparse datasets such as IPUMS-CPS. As for the other algorithms, Algorithm 2 is generally the slowest and most affected, while Algorithm 1 lies in between.

5.4 Accuracy and Privacy Tradeoff

Figure 4 shows the effect of increasing the privacy budget on the relative $L1$ error of the algorithms. Algorithm 3 is consistently the least accurate of the three algorithms, due to its sensitivity being directly proportional to k and inversely proportional to the minimum group size per admissible value (Proposition 3.12). Consequently, for $k = 500$, the noise added for all datasets is larger than that added for the other algorithms; for datasets with small minimum group sizes, such as IPUMS-CPS and Healthcare, this noise is particularly high. The parameter k can be calibrated for each dataset to reduce the noise, as shown in a later experiment, thereby reducing the relative $L1$ error. Algorithm 2 is the most accurate of the three algorithms. This is because its values are much larger in magnitude, so the added DP noise changes the true value only by a small fraction, leading to a smaller relative error. Additionally, the relative error of Algorithm 2 is the most stable across all datasets, since its sensitivity does not depend on the number of tuples (Proposition 3.10), causing the error to be primarily determined by ϵ . Algorithm 1 is less noisy than Algorithm 3, which is consistent with its low sensitivity (Proposition 3.4), but more noisy than Algorithm 2. Moreover, the relative error of Algorithm 1 exhibits large variance between runs, since its true values are small and therefore easily distorted by the added noise. However, for small values of ϵ , the noise dominates the output uniformly across runs, leading to reduced variance in the relative error. Overall, the relative errors of all algorithms steadily decrease as the privacy budget increases, as expected, with the sharpest declining segment occurring up to $\epsilon = 1$.

6 RELATED WORK

Research on fairness in data management intersects several areas, including data valuation, private data pricing, and data inconsistency assessment. Although these directions are important foundations, none develops dedicated measures for data-centric fairness tailored for use in the DP setting, which is the focus of this paper.

A central line of work studies Data Valuation Assessment (DVA) in machine learning, where the goal is to estimate the contribution of individual records or subsets of data to predictive performance [5, 11, 53]. These methods quantify the marginal value of data with respect to accuracy or loss, and emphasize model driven utility rather than fairness or structural properties of the data itself.

Another major body of literature develops formal definitions of fairness [8, 17, 20, 27, 36, 54, 55, 60, 76, 83, 94] and introduces conceptual frameworks to analyze, understand, and mitigate bias in prediction systems [12, 33, 84, 86]. These works largely focus on model behavior or causal interpretations rather than characterizing fairness properties inherent to the dataset itself. As we show in Section 5.2, focusing on dataset unfairness rather than model behavior may be useful in some settings.

Private data valuation methods provide mechanisms for pricing or compensating for the use of sensitive data without addressing fairness concerns [40, 61, 98]. These works typically quantify sensitivity, privacy loss, or economic value but do not analyze disparate impact or structural imbalances in the data.

Inconsistency measures for relational data and integrity constraints [66, 69, 70, 79] offer tools for quantifying violations of integrity constraints. These measures are inspired by long standing work in Knowledge Representation and Logic [42, 43, 49, 56, 57, 75, 90], which develops principles for reasoning under inconsistency. While conceptually related, these methods address logical coherence rather than fairness across protected groups.

Finally, fairness driven data repair methods aim to modify datasets to remove discriminatory patterns as pre- or post-processing solutions [4, 14, 32, 45, 51, 83]. These interventions assume some fairness criterion but do not provide measures designed specifically for assessing fairness within the DP context.

7 CONCLUSION AND LIMITATIONS

We presented a principled framework for quantifying data unfairness under differential privacy, bridging concepts from probabilistic dependence, data repair, and contribution analysis. Our measures satisfy desirable consistency properties, exhibit low sensitivity, and remain tractable under standard DP mechanisms. Our three measures collectively provide complementary perspectives on data level bias. Overall, the framework lays a foundation for a systematic evaluation of fairness in privacy protected data.

While our framework provides a principled foundation for measuring database unfairness under differential privacy, several limitations remain. First, $\mathcal{U}_R^{SAT}$ relies on a MaxSAT-based proxy whose practical instantiation uses a block-wise heuristic. This improves scalability but weakens the estimation ability associated with the original formulation. Even with the heuristic, computing $\mathcal{U}_R^{SAT}$ is slower than computing the other measures, due to the necessity of solving weighted MaxSAT over self-joined chunks. Second, a principled approach for choosing k would make $\mathcal{U}_{TC}$ more user-friendly. Finally, our measures operate purely at the associational level and do not account for causal structure, data collection bias, or label noise, which may also influence fairness assessments.

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A DEMONSTRATING THE FAITHFULNESS OF $\mathcal{U}_{\text{MI}}^{\text{TVD}}$

Figure 6 demonstrates the faithfulness between $\mathcal{U}_{\text{MI}}$ (Figure 6a) and $\mathcal{U}_{\text{MI}}^{\text{Bayes}}$ (Figure 6b) for the Adult, Stackoverflow survey, and Compas datasets (see Example 1.1) with fairness criteria from Table 2, numbered for each dataset. We also plotted $\mathcal{U}_{\text{MI}}^{\text{Bayes}}$ with an offset of $\frac{1}{2}$ (Figure 6c) to make its values positive so the reader could easily see the variation between $\mathcal{U}_{\text{MI}}$ and $\mathcal{U}_{\text{MI}}^{\text{Bayes}}$ across all datasets and criteria.

B THEOREMS AND PROOFS

B.1 Proofs for Section 3

PROOF OF PROPOSITION 3.3. According to the Pinsker’s inequality [22]:

$$\text{TVD}(Pr_D(P, O), Pr_D(P)Pr_D(O))^2 \leq \frac{1}{2} D_{\text{KL}}(Pr_D(P, O) \| Pr_D(P)Pr_D(O)) = \frac{1}{2} \mathcal{U}_{\text{MI}}(P, O)$$

it implies that

$$\mathcal{U}_{\text{MI}}^{\text{TVD}}(P \perp O, D) = 2 \cdot \text{TVD}(Pr_D(P, O), Pr_D(P)Pr_D(O))^2 \leq \mathcal{U}_{\text{MI}}(P, O)$$

From the other side, according to the reverse Pinsker’s inequality, for any two distributions P and Q such that $Q(z) > 0$ for all z it holds that

$$\text{TVD}(P, Q)^2 \geq \frac{1}{2} \cdot \min_z Q(z) \cdot D_{\text{KL}}(P \| Q)$$

Setting $P = Pr_D(P, O)$ and $Q = Pr_D(P)Pr_D(O)$, it implies that

$$\text{TVD}(Pr_D(P, O), Pr_D(P)Pr_D(O))^2 \geq \frac{1}{2} \cdot \alpha \cdot \mathcal{U}_{\text{MI}}(P, O),$$

where $\alpha := \min_{(p,y)} Pr_D(P=p)Pr_D(O=y)$ over the support of $Pr_D(P, O)$.

Multiplying both sides by 2, we get

$$\mathcal{U}_{\text{MI}}^{\text{TVD}}(P \perp O, D) = 2 \cdot \text{TVD}(Pr_D(P, O), Pr_D(P)Pr_D(O))^2 \geq \alpha \cdot \mathcal{U}_{\text{MI}}(P, O)$$

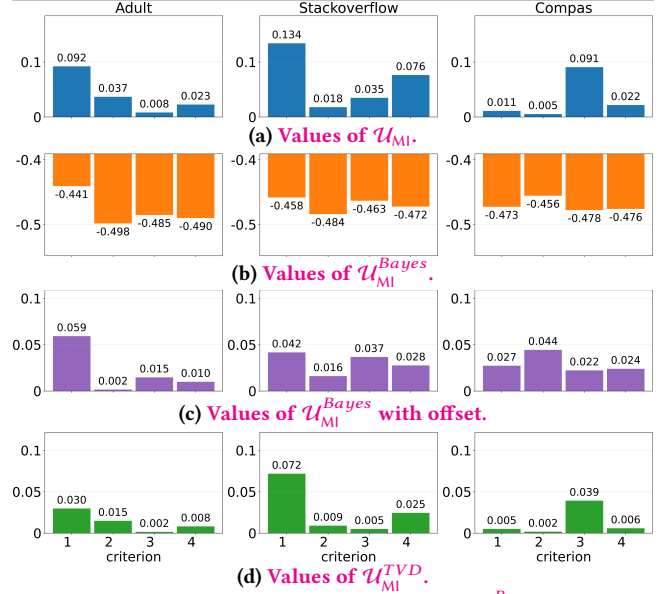


Figure 6: Demonstrating the faithfulness of $\mathcal{U}_{\text{MI}}^{\text{Bayes}}$ and $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ to $\mathcal{U}_{\text{MI}}$ without privacy considerations.

Therefore

$$\alpha \cdot \mathcal{U}_{\text{MI}}(P, O) \leq \mathcal{U}_{\text{MI}}^{\text{TVD}}(P \perp O, D) \leq \mathcal{U}_{\text{MI}}(P, O),$$

where $\alpha := \min_{(p,y)} Pr_D(P=p)Pr_D(O=y)$.

Then, if we define $X = Pr_D(P, O)$, $Y = Pr_D(P)Pr_D(O)$, we would get that the following holds

$$\alpha \cdot \mathcal{U}_{\text{MI}}(P, O) \mathcal{U}_{\text{MI}}^{\text{TVD}}(P \perp O, D) \leq \mathcal{U}_{\text{MI}}(P, O),$$

where $\alpha = \min_{z \in Z, Y(z) > 0} Y(z)$. $\square$

PROOF OF PROPOSITION 3.4 PARTS 1 AND 2. **Positivity:**

By the definition of total variation distance, for any two probability distributions P and Q it holds that

$$\text{TVD}(P, Q) = \frac{1}{2} \sum_x |P(x) - Q(x)| \geq 0$$

and it holds that $\text{TVD}(P, Q) = 0$ iff $P = Q$ by the properties of the absolute value.

For any database D and any fairness criterion $F = P \perp O \mid A \in \mathcal{F}$ where A can be absent, it holds that

$$\begin{aligned} \mathcal{U}_{\text{MI}}^{\text{TVD}}(P \perp O \mid A, D) &= 2 \cdot (\text{TVD}(Pr_D(P, O \mid A), Pr_D(P \mid A)Pr_D(O \mid A)))^2 \\ &\geq \text{TVD}(Pr_D(P, O \mid A), Pr_D(P \mid A)Pr_D(O \mid A)) \\ &\geq 0 \end{aligned}$$

and

$$\begin{aligned} \mathcal{U}_{\text{MI}}^{\text{TVD}}(P \perp O \mid A, D) &= 0 \\ \iff 2 \cdot (\text{TVD}(Pr_D(P, O \mid A), Pr_D(P \mid A)Pr_D(O \mid A)))^2 &= 0 \\ \iff \text{TVD}(Pr_D(P, O \mid A), Pr_D(P \mid A)Pr_D(O \mid A)) &= 0 \\ \iff Pr_D(P, O \mid A) = Pr_D(P \mid A)Pr_D(O \mid A) & \\ \iff F(D) = 1 \end{aligned}$$

We can notice that the above also holds for any a set of fairness criteria $\mathcal{F}$ since $\mathcal{U}_{\text{MI}}^{\text{TVD}}(\mathcal{F}, D) := \sum_{F \in \mathcal{F}} \mathcal{U}_{\text{MI}}^{\text{TVD}}(F, D)$.

Monotonicity:

For any database D and fairness criteria $\mathcal{F}, \mathcal{F}'$ such that $\mathcal{F} \subseteq \mathcal{F}'$, it holds that

$$\mathcal{U}_{\text{MI}}^{\text{TVD}}(\mathcal{F}, D) = \sum_{F \in \mathcal{F}} \mathcal{U}_{\text{MI}}^{\text{TVD}}(F, D) \leq \sum_{F' \in \mathcal{F}'} \mathcal{U}_{\text{MI}}^{\text{TVD}}(F', D) = \mathcal{U}_{\text{MI}}^{\text{TVD}}(\mathcal{F}', D)$$

This follows from the fact that $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ is non-negative (from the Positivity property), and so its sum is monotonic. $\square$

We will first state and prove the range and the sensitivity bounds of $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ for a single fairness criterion, and then we will use them to prove the range and the sensitivity bounds for a set of criteria (Parts 3 and 4 of Proposition 3.4).

LEMMA B.1 ($\mathcal{U}_{\text{MI}}^{\text{TVD}}$ IS SUITABLE FOR DP FOR A SINGLE FAIRNESS CRITERION). *Given a database D of size n , the following holds:*

- (1) *The range of $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ is $[0, 2]$ for a single criterion F .*
- (2) *The sensitivity of $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ for an unconditional fairness criterion $F = P \perp O$ is at most $\frac{12}{n}$ and for a conditional fairness criterion, $F = P \perp O \mid A$, is at most $\frac{16}{n}$.*

PROOF OF LEMMA B.1 PART 1. For any P and Q two probability distributions, it holds that:

$$0 \leq \frac{1}{2} \sum_x |P(x) - Q(x)| \leq \frac{1}{2} (|1 - 0| + |0 - 1|) = 1$$

by the properties of probability. The upper bound is for the case when all probability mass in P is at one outcome, while all probability mass in Q is at the other. In this case, only two terms contribute to the sum, and both contribute 1.

Therefore by the definition of TVD it holds that:

$$0 = 2 \cdot 0^2 \leq 2 \cdot (\text{TVD}(P, Q))^2 \leq 2 \cdot 1^2 = 2$$

And so, by the definition, the range of $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ is $[0, 2]$. $\square$

Now we will state and prove a few lemmas that we will use in a later proof of Lemma B.1 Part 2.

LEMMA B.2 (REVERSE TRIANGLE INEQUALITY). *Given $x, y \in \mathbb{R}$, the following holds*

$$||x| - |y|| \leq |x - y|$$

PROOF. From the triangle inequality,

$$|x| = |(x - y) + y| \leq |x - y| + |y|$$

Rearranging, we get

$$|x| - |y| \leq |x - y|$$

Similarly, by symmetry,

$$|y| = |(y - x) + x| \leq |y - x| + |x| = |x - y| + |x|$$

Rearranging, we get

$$|y| - |x| \leq |x - y|$$

and multiplying both sides by -1 we get

$$-|x - y| \leq -(|y| - |x|) = |x| - |y|$$

Combining the two, we get

$$-|x - y| \leq |x| - |y| \leq |x - y|$$

which by the definition of the absolute value means

$$||x| - |y|| \leq |x - y|$$

$\square$

LEMMA B.3 (SENSITIVITY OF THE SQUARE). *Let $f : \mathcal{D} \rightarrow \mathbb{R}$ be a function with sensitivity Δ_f . Define $g(D) := f(D)^2$. Then the sensitivity of g satisfies:*

$$\Delta_g := \max_{D \sim D'} |g(D) - g(D')| \leq 2 \cdot \Delta_f \cdot \max(\text{Dom}(f))$$

PROOF. Let $D \sim D'$ be neighboring databases. Define $g(D) = f(D)^2$, so

$$\begin{aligned} |g(D) - g(D')| &= |f(D)^2 - f(D')^2| \\ &= |(f(D) - f(D')) \cdot (f(D) + f(D'))| \\ &\leq |f(D) - f(D')| \cdot |f(D) + f(D')| \\ &\leq |f(D) - f(D')| \cdot 2 \cdot \max(\text{Dom}(f)) \\ &\leq \Delta_f \cdot 2 \cdot \max(\text{Dom}(f)) \end{aligned}$$

Taking the maximum over all neighboring databases, we get

$$\Delta_g \leq 2 \cdot \Delta_f \cdot \max(\text{Dom}(f))$$

$\square$

PROOF OF LEMMA B.1 PART 2. **Unconditional case:**

Let there be $D \sim D'$ neighboring databases such that they differ in one tuple by replacement. Without loss of generality, assume that D' contains one more occurrence of $t'' = (p_{t''}, y_{t''})$ and one less occurrence of $t' = (p_{t'}, y_{t'})$ than D .

From Definition 3.2, by the triangle inequality and Lemma B.2, we have the following

$$|\text{TVD}(Pr_{D'}(P, O), Pr_{D'}(P)Pr_{D'}(O)) - \text{TVD}(Pr_D(P, O), Pr_D(P)Pr_D(O))| \quad (1)$$

$$= \left| \frac{1}{2} \sum_{p \in P, y \in O} |Pr_{D'}(P = p, O = y) - Pr_{D'}(P = p)Pr_{D'}(O = y)| - \frac{1}{2} \sum_{p \in P, y \in O} |Pr_D(P = p, O = y) - Pr_D(P = p)Pr_D(O = y)| \right| \quad (2)$$

$$\leq \left| \frac{1}{2} \sum_{p \in P, y \in O} \left(Pr_{D'}(P = p, O = y) - Pr_{D'}(P = p)Pr_{D'}(O = y) - Pr_D(P = p, O = y) + Pr_D(P = p)Pr_D(O = y) \right) \right| \quad (3)$$

$$= \frac{1}{2} \left| \sum_{p \in P, y \in O} \left(Pr_{D'}(P = p, O = y) - Pr_D(P = p, O = y) \right) + \sum_{p \in P, y \in O} \left(Pr_D(P = p)Pr_D(O = y) - Pr_{D'}(P = p)Pr_{D'}(O = y) \right) \right| \quad (4)$$

$$\leq \frac{1}{2} \left| \sum_{p \in P, y \in O} Pr_{D'}(P = p, O = y) - Pr_D(P = p, O = y) \right| + \frac{1}{2} \left| \sum_{p \in P, y \in O} Pr_{D'}(P = p)Pr_{D'}(O = y) - Pr_D(P = p)Pr_D(O = y) \right| \quad (5)$$

$$\leq \frac{1}{2} \sum_{p \in P, y \in O} |Pr_{D'}(P = p, O = y) - Pr_D(P = p, O = y)| + \frac{1}{2} \sum_{p \in P, y \in O} |Pr_{D'}(P = p)Pr_{D'}(O = y) - Pr_D(P = p)Pr_D(O = y)| \quad (6)$$

The transition from (2) to (3) is due to Lemma B.2, the transitions from (4) to (5) and from (5) to (6) are due to triangle inequality.

Looking at the first part of the sum separately, we get

$$\begin{aligned} &\frac{1}{2} \sum_{p \in P, y \in O} |Pr_{D'}(P = p, O = y) - Pr_D(P = p, O = y)| \\ &\leq \frac{1}{2} \left(|Pr_{D'}(P = p_{t'}, O = y_{t'}) - Pr_D(P = p_{t'}, O = y_{t'})| \right. \\ &\quad \left. + |Pr_{D'}(P = p_{t''}, O = y_{t''}) - Pr_D(P = p_{t''}, O = y_{t''})| \right) \\ &\leq \frac{1}{2} \cdot \left(\frac{1}{n} + \frac{1}{n} \right) \\ &= \frac{1}{n} \end{aligned}$$

Looking at the second part of the sum, we get

$$\frac{1}{2} \sum_{p \in P, y \in O} |Pr_{D'}(P=p) Pr_{D'}(O=y) - Pr_D(P=p) Pr_D(O=y)| \quad (7)$$

$$= \frac{1}{2} \sum_{p \in P, y \in O} |Pr_{D'}(P=p) Pr_{D'}(O=y) + Pr_D(P=p) Pr_{D'}(O=y) - Pr_D(P=p) Pr_D(O=y) - Pr_D(P=p) Pr_D(O=y)| \quad (8)$$

$$= \frac{1}{2} \sum_{p \in P, y \in O} |(Pr_{D'}(P=p) - Pr_D(P=p)) Pr_{D'}(O=y) + Pr_D(P=p) (Pr_{D'}(O=y) - Pr_D(O=y))| \quad (9)$$

$$\leq \frac{1}{2} \sum_{p \in P, y \in O} (|Pr_{D'}(P=p) - Pr_D(P=p)| Pr_{D'}(O=y) + Pr_D(P=p) |Pr_{D'}(O=y) - Pr_D(O=y)|) \quad (10)$$

$$= \frac{1}{2} \left(\sum_{p \in P} |Pr_{D'}(P=p) - Pr_D(P=p)| \sum_{y \in O} Pr_{D'}(O=y) + \sum_{y \in O} |Pr_{D'}(O=y) - Pr_D(O=y)| \sum_{p \in P} Pr_D(P=p) \right) \quad (11)$$

$$= \frac{1}{2} \left(\sum_{p \in P} |Pr_{D'}(P=p) - Pr_D(P=p)| \cdot 1 + \sum_{y \in O} |Pr_{D'}(O=y) - Pr_D(O=y)| \cdot 1 \right) \quad (12)$$

$$= \frac{1}{2} \sum_{p \in P} |Pr_{D'}(P=p) - Pr_D(P=p)| + \frac{1}{2} \sum_{y \in O} |Pr_{D'}(O=y) - Pr_D(O=y)| \quad (13)$$

where the transition from (9) to (10) is due to applying the triangle inequality, the transition from (11) to (12) is due to the fact that the sum of probabilities for all the values is 1.

In particular,

$$\begin{aligned} & \frac{1}{2} \sum_{p \in P} |Pr_{D'}(P=p) - Pr_D(P=p)| \\ & \leq \frac{1}{2} \cdot \left(\frac{1}{n} + \frac{1}{n} \right) \\ & = \frac{1}{n} \end{aligned}$$

A similar expression can be achieved by swapping P and p for O and y , respectively.

Putting it all together,

$$\begin{aligned} & |\text{TVD}(Pr_{D'}(P, O), Pr_{D'}(P) Pr_{D'}(O)) - \text{TVD}(Pr_D(P, O), Pr_D(P) Pr_D(O))| \\ & \leq \frac{1}{2} \sum_{p \in P, y \in O} |Pr_{D'}(p, y) - Pr_D(p, y)| \\ & \quad + \frac{1}{2} \sum_{p \in P, y \in O} |Pr_{D'}(p) Pr_{D'}(y) - Pr_D(p) Pr_D(y)| \\ & \leq \frac{1}{n} + \frac{1}{2} \sum_{p \in P} |Pr_{D'}(p) - Pr_D(p)| + \frac{1}{2} \sum_{y \in O} |Pr_{D'}(y) - Pr_D(y)| \\ & \leq \frac{3}{n} \end{aligned}$$

Taking the maximum over all neighboring databases $D \sim D'$, we get

$$\Delta_{\text{TVD}} \leq \frac{3}{n}$$

From Lemma B.3, it holds that

$$\begin{aligned} \Delta_{\mathcal{U}_{\text{MI}}^{\text{TVD}}} &= \Delta_{2\text{TVD}^2} \\ &= 2 \cdot \Delta_{\text{TVD}^2} \\ &\leq 2 \cdot 2 \cdot \Delta_{\text{TVD}} \cdot \max(\text{Dom}(\text{TVD})) \\ &\stackrel{\text{TVD} \leq 1}{\leq} 4 \cdot \frac{3}{n} \cdot 1 \\ &= \frac{12}{n} \end{aligned}$$

Therefore, the sensitivity bound for the unconditional case is $\frac{12}{n}$.

Conditional case:

Similarly, assume that D' contains one more occurrence of $t'' = (p_{t''}, y_{t''}, a_{t''})$ and one less occurrence of $t' = (p_{t'}, y_{t'}, a_{t'})$ than D .

Denote $f_D(a) := \mathcal{U}_{\text{MI}}^{\text{TVD}}(P \perp O \mid A=a, D)$, and similarly $f_{D'}(a)$. Then from Definition 3.2 it holds that

$$|\mathcal{U}_{\text{MI}}^{\text{TVD}}(P \perp O \mid A, D') - \mathcal{U}_{\text{MI}}^{\text{TVD}}(P \perp O \mid A, D)| \quad (14)$$

$$= \left| \sum_a (Pr_{D'}(A=a) f_{D'}(a) - Pr_D(A=a) f_D(a)) \right| \quad (15)$$

$$\leq \sum_a |Pr_{D'}(A=a) f_{D'}(a) - Pr_D(A=a) f_D(a)| \quad (16)$$

$$= \sum_a \left| (Pr_{D'}(A=a) - Pr_D(A=a)) f_{D'}(a) + (Pr_D(A=a) (f_{D'}(a) - f_D(a))) \right| \quad (17)$$

$$\leq \sum_a \left| (Pr_{D'}(A=a) - Pr_D(A=a)) f_{D'}(a) \right| + |Pr_D(A=a) (f_{D'}(a) - f_D(a))| \quad (18)$$

$$\leq \sum_a |Pr_{D'}(A=a) - Pr_D(A=a)| |f_{D'}(a)| + \sum_a Pr_D(A=a) |f_{D'}(a) - f_D(a)|, \quad (19)$$

where the transitions are due to:

- (15) $\rightarrow$ (17): triangle inequality.
- (17) $\rightarrow$ (18): triangle inequality.
- (18) $\rightarrow$ (19): $xy \leq |x||y|$ for $x, y \in \mathbb{R}$.

Looking at the first sum separately, because only $a \in \{a_{t'}, a_{t''}\}$ change, and since $|f_{D'}(a)| \leq 2$ by the Range property from Proposition 3.4 proved earlier, then

$$\sum_a |Pr_{D'}(A=a) - Pr_D(A=a)| |f_{D'}(a)| \leq \left(\frac{1}{n} + \frac{1}{n} \right) \cdot 2 = \frac{4}{n}$$

Looking at the second sum, by the unconditional case and by the Lemma B.3, for a fixed a with $n_a = |\{t \in D \mid t[A] = a\}|$, it holds that the sensitivity of $2 \cdot \text{TVD}^2$ is at most $\frac{12}{n_a}$ when both removing t' and adding t'' are included in the computation. So, $|f_{D'}(a_{t'}) - f_D(a_{t'})| = \frac{1}{2} \cdot \frac{12}{n_{a_{t'}}}$, and similarly for t'' . Therefore

$$\begin{aligned} & \sum_a Pr_D(A=a) |f_{D'}(a) - f_D(a)| \\ &= Pr_D(A=a_{t'}) |f_{D'}(a_{t'}) - f_D(a_{t'})| + Pr_D(A=a_{t''}) |f_{D'}(a_{t''}) - f_D(a_{t''})| \\ &\leq \frac{n_{a_{t'}}}{n} \cdot \frac{6}{n_{a_{t'}}} + \frac{n_{a_{t''}}}{n} \cdot \frac{6}{n_{a_{t''}}} \\ &= \frac{12}{n} \end{aligned}$$

Putting it all together,

$$\Delta_{\mathcal{U}_{\text{MI}}^{\text{TVD}}} = \max_{D' \sim D} |\mathcal{U}_{\text{MI}}^{\text{TVD}}(P \perp O \mid A, D') - \mathcal{U}_{\text{MI}}^{\text{TVD}}(P \perp O \mid A, D)| \leq \frac{4}{n} + \frac{12}{n} = \frac{16}{n}$$

Therefore, the sensitivity bound for the conditional case is $\frac{16}{n}$. $\square$

PROOF OF PROPOSITION 3.4 PARTS 3 AND 4. Range:

By Definition 3.2 and by Lemma B.1, it holds that

$$\mathcal{U}_{\text{MI}}^{\text{TVD}}(\mathcal{F}, D) = \sum_{F \in \mathcal{F}} \mathcal{U}_{\text{MI}}^{\text{TVD}}(F, D) \geq \sum_{F \in \mathcal{F}} 0 = 0$$

and

$$\mathcal{U}_{\text{MI}}^{\text{TVD}}(\mathcal{F}, D) = \sum_{F \in \mathcal{F}} \mathcal{U}_{\text{MI}}^{\text{TVD}}(F, D) \leq \sum_{F \in \mathcal{F}} 2 = 2|\mathcal{F}|$$

Sensitivity:

From Lemma B.1, the sensitivity for a single fairness criterion F is at most

$$\Delta_{\mathcal{U}_{\text{MI}}^{\text{TVD}}(F, D)} \leq \frac{16}{n}$$

Therefore, for $D' \sim D$, using the triangle inequality and linearity of the sum, it holds that

$$\begin{aligned} |\mathcal{U}_{\text{MI}}^{TV D}(\mathcal{F}, D') - \mathcal{U}_{\text{MI}}^{TV D}(\mathcal{F}, D)| &= \left| \sum_{F \in \mathcal{F}} (\mathcal{U}_{\text{MI}}^{TV D}(F, D')) - \sum_{F \in \mathcal{F}} (\mathcal{U}_{\text{MI}}^{TV D}(F, D)) \right| \\ &\leq \sum_{F \in \mathcal{F}} |\mathcal{U}_{\text{MI}}^{TV D}(F, D') - \mathcal{U}_{\text{MI}}^{TV D}(F, D)| \\ &\leq \sum_{F \in \mathcal{F}} \Delta_{\mathcal{U}_{\text{MI}}^{TV D}(F, D)} \\ &\leq \frac{16|\mathcal{F}|}{n} \end{aligned}$$

□

PROPOSITION B.4 ($\mathcal{U}_R$ SATISFIES THE DESIRED PROPERTIES).

- (1) $\mathcal{U}_R$ satisfies the Positivity property.
- (2) $\mathcal{U}_R$ satisfies the Monotonicity property.
- (3) The range of $\mathcal{U}_R$ is $[0, n]$ for a set of criteria $\mathcal{F}$.
- (4) The sensitivity of $\mathcal{U}_R$ is 1 for a set of criteria $\mathcal{F}$.

PROOF OF PROPOSITION B.4 PARTS 1 AND 2. **Positivity:**

The positivity property holds, as Definition 3.5 defines $\mathcal{U}_R(\mathcal{F}, D) = |D \div D_R|$ with an absolute value.

Monotonicity:

For monotonicity, assume that $\mathcal{F} \subseteq \hat{\mathcal{F}}$ and that D_R and $\hat{D}_R$ are the minimum repaired databases w.r.t. D and $\mathcal{F}, \hat{\mathcal{F}}$ respectively.

By definition $\hat{D}_R$ satisfies $\hat{\mathcal{F}}$ and thus in particular satisfies $\mathcal{F}$. Hence, $|D \div \hat{D}_R|$ can only be equal or larger than $|D \div D_R|$, since by our definition D_R is the database that **minimizes** the symmetric difference with D and satisfies $\mathcal{F}$. Therefore, $\mathcal{U}_R(\mathcal{F}, D) = |D \div D_R| \leq |D \div \hat{D}_R| = \mathcal{U}_R(\hat{\mathcal{F}}, D)$. □

We will first state and prove the range and the sensitivity bounds of $\mathcal{U}_R$ for a single fairness criterion, and then we will use them to prove the range and the sensitivity bounds for a set of criteria (Parts 3 and 4 of Proposition B.4).

LEMMA B.5 ($\mathcal{U}_R$ IS SUITABLE FOR DP FOR A SINGLE FAIRNESS CRITERION). *Given a database D of size n , the following holds:*

- (1) The range of $\mathcal{U}_R(F, D)$ is $[0, n]$ for a single criterion F .
- (2) The sensitivity of $\mathcal{U}_R$ is at most 1 for a single criterion F .

PROOF. **Range:**

It is immediate that $\text{Range}(\mathcal{U}_R(F, D)) = [0, n]$ because $\mathcal{U}_R(F, D) \geq 0$ from the Positivity property, and $\mathcal{U}_R(F, D) \leq n$ because in the most extreme case, deleting all tuples in D will create the empty database that satisfies F .

Sensitivity:

Given F and $D \sim D'$ where without loss of generality $D' = D \setminus \{t'\} \cup \{t''\}$ for some tuples t' and t'' , we have $\mathcal{U}_R(F, D) = |D \div D_R|$, where D_R is the database that satisfies F with the minimum number of deletions and additions.

Since D' differs from D by replacing a single tuple t' with t'' , we can obtain the repair process for D' from the repair process for D as follows:

- If t'' interferes with F , replace t'' with t' to obtain D , and then do the repair for D . In this case $\mathcal{U}_R(F, D') = \mathcal{U}_R(F, D) + 1$.
- Otherwise, t'' does not interfere with F , and so in this case $\mathcal{U}_R(F, D') = \mathcal{U}_R(F, D)$.

Therefore, $\Delta_{\mathcal{U}_R(F, D)} = \max_{D' \sim D} |\mathcal{U}_R(F, D') - \mathcal{U}_R(F, D)| \leq 1$, and the sensitivity of $\mathcal{U}_R$ is at most 1 for a single criterion F . □

PROOF OF PROPOSITION B.4 PARTS 3 AND 4. **Range:**

Again, $\text{Range}(\mathcal{U}_R(\mathcal{F}, D)) = [0, n]$ because $\mathcal{U}_R(\mathcal{F}, D) \geq 0$ from the Positivity property, and $\mathcal{U}_R(\mathcal{F}, D) \leq n$ because in the most extreme case, deleting all tuples in D will create the empty database that satisfies $\mathcal{F}$.

Sensitivity:

Let there be a database D_R with the smallest number of removed and added tuples with respect to D that satisfies all the fairness criteria $\mathcal{F}$. Then

$$\mathcal{U}_R(\mathcal{F}, D') \leq |D' \div D_R| \leq |D \div D_R| + 1 = \mathcal{U}_R(\mathcal{F}, D) + 1$$

From another side, let there be a database D'_R with the smallest number of removed and added tuples with respect to D' that satisfies all the fairness criteria $\mathcal{F}$. Then

$$\mathcal{U}_R(\mathcal{F}, D) \leq |D \div D'_R| \leq |D' \div D'_R| + 1 = \mathcal{U}_R(\mathcal{F}, D') + 1$$

Therefore

$$\Delta_{\mathcal{U}_R(\mathcal{F}, D)} = \max_{D' \sim D} |\mathcal{U}_R(\mathcal{F}, D') - \mathcal{U}_R(\mathcal{F}, D)| \leq 1$$

□

PROOF OF PROPOSITION 3.12 PARTS 1 AND 2. **Positivity:**

Recall that by Definition 3.11, given a single fairness criterion F , $\mathcal{U}_{\text{TC}}$ is an absolute value. Since its aggregate form is a sum $\mathcal{U}_{\text{TC}}$ is always non-negative and its value is 0 only if the databases satisfies all the given fairness criteria in $\mathcal{F}$.

Monotonicity:

Monotonicity follows directly from positivity. □

We will first state and prove the range and the sensitivity bounds of $\mathcal{U}_{\text{TC}}$ for a single fairness criterion, and then we will use them to prove the range and the sensitivity bounds for a set of criteria (Parts 3 and 4 of Proposition 3.12).

LEMMA B.6 ($\mathcal{U}_{\text{TC}}$ IS SUITABLE FOR DP FOR A SINGLE FAIRNESS CRITERION). *Given a database D and a parameter $k \in \mathbb{N}$, the following holds:*

- (1) The range of $\mathcal{U}_{\text{TC}}(F, D)$ is $[0, \min\{\frac{k}{4}, 2\}]$ for a single criterion F .
- (2) The sensitivity of $\mathcal{U}_{\text{TC}}(F, D)$ is at most $\frac{3k}{n}$ for an unconditional criterion F , and, assuming that $|\{t \in D \mid t[A] = a\}| \geq 2$ for every $a \in A$, is at most $\frac{7k}{n}$ for a conditional one.

PROOF OF LEMMA B.6 PART 1. By Definition 3.11, given a single fairness criterion F , $\mathcal{U}_{\text{TC}}$ is an absolute value. Therefore, from the properties of the absolute value, $\mathcal{U}_{\text{TC}}(F, D) \geq 0$.

For every $t = (p, y, a) \in \text{top} - k$, define:

$$q_t = \Pr_D(P = t[P], O = t[O] \mid A = t[A]),$$

$$p_t = \Pr_D(P = t[P] \mid A = t[A]),$$

$$r_t = \Pr_D(O = t[O] \mid A = t[A]).$$

Then by the Fréchet inequalities [35], it holds that $\max\{0, p_t + r_t - 1\} \leq q_t \leq \min\{p_t, r_t\}$, and so for the convex function $q_t \rightarrow |q_t - p_t r_t|$, we get that:

$$|q_t - p_t r_t| \leq \max\{|\min\{p_t, r_t\} - p_t r_t|, |\max\{0, p_t + r_t - 1\} - p_t r_t|\}$$

Without loss of generality, assume that $p_t \leq r_t$ (the other case is symmetric). Then:

$$|\min\{p_t, r_t\} - p_t r_t| = |p_t - p_t r_t| = p_t(1 - r_t) \leq r_t(1 - r_t) \leq \max_{r_t = \frac{1}{2}} \frac{1}{4}$$

For the other side there are two cases. If $p_t + r_t < 1$, then:

$$|\max\{0, p_t + r_t - 1\} - p_t r_t| = p_t r_t \leq \left(\frac{p_t + r_t}{2}\right)^2 \leq \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

Otherwise, $p_t + r_t \geq 1$, and:

$$\begin{aligned} |\max\{0, p_t + r_t - 1\} - p_t r_t| &= |(p_t + r_t - 1) - p_t r_t| \\ &= |-(1 - p_t)(1 - r_t)| \\ &= (1 - p_t)(1 - r_t) \\ &\leq \max_{p_t=r_t=\frac{1}{2}} \frac{1}{4} \end{aligned}$$

Therefore, we got that $|q_t - p_t r_t| \leq \max\{\frac{1}{4}, \frac{1}{4}\} = \frac{1}{4}$, and so

$$\begin{aligned} \mathcal{U}_{TC}(F, D) &= \sum_{t=(p,y,a) \in \text{top-}k} \mathcal{MD}(P \perp O \mid A, D, t) \\ &= \sum_{t=(p,y,a) \in \text{top-}k} |Pr_D(A = t[A]) Pr_D(P = t[P], O = t[O] \mid A = t[A]) \\ &\quad - Pr_D(P = t[P] \mid A = t[A]) Pr_D(O = t[O] \mid A = t[A])| \\ &\leq \sum_{t=(p,y,a) \in \text{top-}k} 1 \cdot |Pr_D(P = t[P], O = t[O] \mid A = t[A]) \\ &\quad - Pr_D(P = t[P] \mid A = t[A]) Pr_D(O = t[O] \mid A = t[A])| \\ &= \sum_{t \in \text{top-}k} |q_t - p_t r_t| \\ &\leq \sum_{t \in \text{top-}k} \frac{1}{4} \\ &\leq \frac{k}{4} \end{aligned}$$

Additionally, from the proof of Lemma B.1, it holds that

$$\begin{aligned} \mathcal{U}_{TC}(F, D) &\leq \sum_{t \in \text{top-}k} |q_t - p_t r_t| \\ &\leq \sum_{t \in D} |q_t - p_t r_t| \\ &= 2 \cdot \text{TVD}(q_t, p_t r_t) \\ &\leq 2 \cdot 1 \\ &\leq 2 \end{aligned}$$

Therefore:

$$\mathcal{U}_{TC}(F, D) \leq \min\{\frac{k}{4}, 2\}$$

□

Now we will state and prove a few lemmas that we will use in a later proof of Lemma B.6 Part 2.

LEMMA B.7 (SENSITIVITY OF THE UNCONDITIONAL EMPIRICAL PROBABILITIES). *Given a database D of size n such that the schema of D is a superset of $P \cup O$, the sensitivity of the unconditional empirical probabilities $Pr_D(P, O), Pr_D(P), Pr_D(O)$ in D is at most $\frac{1}{n}$.*

PROOF. We will prove this claim for the **joint** empirical probabilities. The same calculation can be done for the marginal empirical probabilities.

Let there be $D \sim D'$ neighboring databases that differ by one tuple $t' = (p_{t'}, y_{t'})$. Assume without loss of generality that $t' \in D'$ and $t' \notin D$.

For any (p, y) , it holds that

$$Pr_D(P = p, O = y) = \frac{|\{t \in D \mid t[P] = p, t[O] = y\}|}{n}$$

For $Pr_{D'}$ we divide into two cases. For any $(p, y) \neq (p_{t'}, y_{t'})$, it holds that

$$\begin{aligned} Pr_{D'}(P = p, O = y) &= \frac{|\{t \in D' \mid t[P] = p, t[O] = y\}|}{n} \\ &= \frac{|\{t \in D \mid t[P] = p, t[O] = y\}|}{n} \\ &\leq \frac{|\{t \in D \mid t[P] = p, t[O] = y\}| + 1}{n} \end{aligned}$$

Otherwise, for $(p, y) = (p_{t'}, y_{t'})$, and

$$\begin{aligned} Pr_{D'}(P = p_{t'}, O = y_{t'}) &= \frac{|\{t \in D' \mid t[P] = p_{t'}, t[O] = y_{t'}\}|}{n} \\ &= \frac{|\{t \in D \mid t[P] = p_{t'}, t[O] = y_{t'}\}| + 1}{n} \end{aligned}$$

Therefore, for any (p, y) , we obtain that

$$\begin{aligned} |Pr_{D'}(P = p, O = y) - Pr_D(P = p, O = y)| \\ \leq \left| \frac{|\{t \in D \mid t[P] = p, t[O] = y\}| + 1}{n} - \frac{|\{t \in D \mid t[P] = p, t[O] = y\}|}{n} \right| \\ = \frac{1}{n} \end{aligned}$$

And so

$$\Delta Pr_D \leq \frac{1}{n}$$

□

Now we will state and prove a few lemmas that we will use in a later proof of Lemma B.6 Part 2.

LEMMA B.8 (SENSITIVITY OF THE CONDITIONAL EMPIRICAL PROBABILITIES). *Given a database D of size n such that the schema of D is a superset of $P \cup O \cup A$, and assuming that $|\{t \in D \mid t[A] = a\}| \geq 2$ for every $a \in A$, the sensitivity of the conditional empirical probabilities $Pr_D(P, O \mid A), Pr_D(P \mid A), Pr_D(O \mid A)$ in D is at most $\max_a \frac{1}{|\{t \in D \mid t[A] = a\}| - 1}$.*

PROOF. We will prove this claim for the **joint** empirical probabilities. The same calculation can be done for the marginal empirical probabilities.

Let there be $D \sim D'$ neighboring databases such that they differ in one tuple by replacement. Without loss of generality, assume that D' contains one more occurrence of t'' and one less occurrence of t' than D .

There are three cases. For any (p, y, a) such that $a \notin \{a_{t'}, a_{t''}\}$, it holds that

$$Pr_{D'}(P = p, O = y \mid A = a) = \frac{|\{t \in D' \mid t[P] = p, t[O] = y, t[A] = a\}|}{|\{t \in D' \mid t[A] = a\}|} \quad (20)$$

$$= \frac{|\{t \in D \mid t[P] = p, t[O] = y, t[A] = a\}|}{|\{t \in D \mid t[A] = a\}|} = Pr_D(P = p, O = y \mid A = a) \quad (21)$$

For any (p, y, a) such that $a = a_{t'}$, this bucket of tuples loses one tuple. Therefore

$$Pr_{D'}(P = p, O = y \mid A = a) = \frac{|\{t \in D \mid t[P] = p, t[O] = y, t[A] = a_{t'}\}| - 1}{|\{t \in D \mid t[A] = a_{t'}\}| - 1}$$

Similarly, for any (p, y, a) such that $a = a_{t''}$, it holds that

$$Pr_{D'}(P = p, O = y \mid A = a) = \frac{|\{t \in D \mid t[P] = p, t[O] = y, t[A] = a_{t''}\}| + 1}{|\{t \in D \mid t[A] = a_{t''}\}| + 1}$$

And so, for any (p, y, a) , we obtain that

$$|Pr_{D'}(P = p, O = y \mid A = a) - Pr_D(P = p, O = y \mid A = a)| \quad (22)$$

$$\leq \begin{cases} \frac{1}{|\{t \in D \mid t[A] = a_{t'}\}| - 1}, & \text{if } a = a_{t'}, \\ \frac{1}{|\{t \in D \mid t[A] = a_{t''}\}| + 1}, & \text{if } a = a_{t''}, \\ 0, & \text{otherwise.} \end{cases} \quad (23)$$

Therefore

$$\Delta_{Pr_D} \leq \max_a \frac{1}{|\{t \in D \mid t[A] = a\}| - 1}$$

□

LEMMA B.9. *Given a database D of size n such that the schema of D is a superset of $P \cup O \cup A$, the sensitivity of the marginal difference is at most $\frac{3}{n}$ in the unconditional case and, assuming that $|\{t \in D \mid t[A] = a\}| \geq 2$ for every $a \in A$, is at most $\frac{7}{n}$ in the conditional case.*

PROOF. Let there be $D \sim D'$ neighboring databases such that they differ in one tuple by replacement. Without loss of generality, assume that D' contains one more occurrence of t'' and one less occurrence of t' than D .

Unconditional case:

For any (p, y) , it holds that

$$|\mathcal{MD}(P \perp O, D', (p, y)) - \mathcal{MD}(P \perp O, D, (p, y))| \quad (24)$$

$$= \left| \left| Pr_{D'}(P = p, O = y) - Pr_{D'}(P = p) Pr_{D'}(O = y) \right| - \left| Pr_D(P = p, O = y) - Pr_D(P = p) Pr_D(O = y) \right| \right| \quad (25)$$

$$\leq \left| \left(Pr_{D'}(P = p, O = y) - Pr_{D'}(P = p) Pr_{D'}(O = y) \right) - \left(Pr_D(P = p, O = y) - Pr_D(P = p) Pr_D(O = y) \right) \right| \quad (26)$$

$$= \left| \left(Pr_{D'}(P = p, O = y) - Pr_D(P = p, O = y) \right) - \left(Pr_{D'}(P = p) Pr_{D'}(O = y) - Pr_D(P = p) Pr_D(O = y) \right) \right| \quad (27)$$

$$= \left| \left[Pr_{D'}(P = p, O = y) - Pr_D(P = p, O = y) \right] - \left[Pr_{D'}(P = p) (Pr_{D'}(O = y) - Pr_D(O = y)) + Pr_D(O = y) (Pr_{D'}(P = p) - Pr_D(P = p)) \right] \right| \quad (28)$$

$$\leq \left| Pr_{D'}(P = p, O = y) - Pr_D(P = p, O = y) \right| + \left| Pr_{D'}(P = p) (Pr_{D'}(O = y) - Pr_D(O = y)) \right| + \left| Pr_D(O = y) (Pr_{D'}(P = p) - Pr_D(P = p)) \right| \quad (29)$$

$$\leq \frac{1}{n} + \left| Pr_{D'}(P = p) \right| \cdot \frac{1}{n} + \left| Pr_D(O = y) \right| \cdot \frac{1}{n} \quad (30)$$

$$\leq \frac{3}{n}, \quad (31)$$

where the transitions are due to:

- (25) $\rightarrow$ (26): Lemma B.2.
- (27) $\rightarrow$ (28): $ab - cd = a(b - d) + d(a - c)$.
- (28) $\rightarrow$ (29): triangle inequality.
- (29) $\rightarrow$ (30): Lemma B.7.
- (30) $\rightarrow$ (31): $Pr_{D'}(P = p) \leq 1, Pr_D(O = y) \leq 1$.

And so

$$\Delta_{\mathcal{MD}(F,D,t)} \leq \frac{3}{n}$$

Conditional case:

By the definition of conditional $\mathcal{MD}$ it holds that

$$|\mathcal{MD}(P \perp O \mid A, D', t) - \mathcal{MD}(P \perp O \mid A, D, t)| \quad (32)$$

$$= \left| \left| Pr_{D'}(A = a) \left| Pr_{D'}(P = p, O = y \mid A = a) - Pr_{D'}(P = p \mid A = a) Pr_{D'}(O = y \mid A = a) \right| - Pr_D(A = a) \left| Pr_D(P = p, O = y \mid A = a) - Pr_D(P = p \mid A = a) Pr_D(O = y \mid A = a) \right| \right| \right| \quad (33)$$

$$\leq \left| Pr_{D'}(A = a) \left(Pr_{D'}(P = p, O = y \mid A = a) - Pr_{D'}(P = p \mid A = a) Pr_{D'}(O = y \mid A = a) \right) - Pr_D(A = a) \left(Pr_D(P = p, O = y \mid A = a) - Pr_D(P = p \mid A = a) Pr_D(O = y \mid A = a) \right) \right| \quad (34)$$

$$\leq \left| Pr_{D'}(A = a) - Pr_D(A = a) \right| \cdot \left| Pr_{D'}(P = p, O = y \mid A = a) - Pr_{D'}(P = p \mid A = a) Pr_{D'}(O = y \mid A = a) \right| + Pr_D(A = a) \cdot \left| \left(Pr_{D'}(P = p, O = y \mid A = a) - Pr_{D'}(P = p \mid A = a) Pr_{D'}(O = y \mid A = a) \right) - \left(Pr_D(P = p, O = y \mid A = a) - Pr_D(P = p \mid A = a) Pr_D(O = y \mid A = a) \right) \right|, \quad (35)$$

where the transition from (33) to (34) is due to Lemma B.2, and the transition from (34) to (35) is due to the fact that $|a'b' - ab| \leq |a' - a| |b'| + |a| |b' - b|$.

We will start by bounding the first summand. From Lemma B.7, $\Delta_{Pr_D(A)} = \frac{1}{n}$. In addition, from the proof of Lemma B.6 Part 1, for every (p, y, a) ,

$$\left| Pr_D(P = p, O = y \mid A = a) - Pr_D(P = p \mid A = a) Pr_D(O = y \mid A = a) \right| \leq \frac{1}{4}$$

Therefore

$$\begin{aligned} & \left| Pr_{D'}(A = a) - Pr_D(A = a) \right| \cdot \left| Pr_{D'}(P = p, O = y \mid A = a) - Pr_{D'}(P = p \mid A = a) Pr_{D'}(O = y \mid A = a) \right| \\ & \leq \frac{1}{n} \cdot \frac{1}{4} \\ & = \frac{1}{4n} \end{aligned}$$

Now we will bound the second summand. Denote:

$$p' := Pr_{D'}(P = p, O = y \mid A = a),$$

$$q' := Pr_{D'}(P = p \mid A = a),$$

$$r' := Pr_{D'}(O = y \mid A = a),$$

$$p := Pr_D(P = p, O = y \mid A = a),$$

$$q := Pr_D(P = p \mid A = a),$$

$$r := Pr_D(O = y \mid A = a).$$

Then it holds that

$$Pr_D(A = a) \cdot \left| \left(Pr_{D'}(P = p, O = y \mid A = a) - Pr_{D'}(P = p \mid A = a) Pr_{D'}(O = y \mid A = a) \right) - \left(Pr_D(P = p, O = y \mid A = a) - Pr_D(P = p \mid A = a) Pr_D(O = y \mid A = a) \right) \right| \quad (36)$$

$$= Pr_D(A = a) \cdot \left| (p' - q'r') - (p - qr) \right| \quad (37)$$

$$= Pr_D(A = a) \cdot \left| (p' - p) - (q'r' - qr) \right| \quad (38)$$

$$\leq Pr_D(A = a) \cdot (|p' - p| + |q'r' - qr|) \quad (39)$$

$$\leq Pr_D(A = a) \cdot (|p' - p| + |q'(r' - r) + r(q' - q)|) \quad (40)$$

$$\leq Pr_D(A = a) \cdot (|p' - p| + |q'(r' - r)| + |r(q' - q)|) \quad (41)$$

$$\leq Pr_D(A = a) \cdot (|p' - p| + |q'(r' - r)| + |r(q' - q)|) \quad (42)$$

$$\leq Pr_D(A = a) \cdot (|p' - p| + |r' - r| + |q' - q|) \quad (43)$$

$$\leq Pr_D(A = a) \cdot 3 \cdot \frac{1}{|\{t \in D \mid t[A] = a\}| - 1} \quad (44)$$

$$= \frac{|\{t \in D \mid t[A] = a\}|}{n} \cdot \frac{3}{|\{t \in D \mid t[A] = a\}| - 1} \quad (45)$$

$$= \frac{3}{n} \cdot \frac{|\{t \in D \mid t[A] = a\}|}{|\{t \in D \mid t[A] = a\}| - 1} \quad (46)$$

$$\leq \frac{3}{n} \cdot 2 \quad (47)$$

$$= \frac{6}{n}, \quad (48)$$

where the transitions are due to:

- (39) $\rightarrow$ (40): triangle inequality.
- (40) $\rightarrow$ (41): $ab - cd = a(b - d) + d(a - c)$.
- (41) $\rightarrow$ (42): triangle inequality.
- (42) $\rightarrow$ (43): $q' \leq 1, r \leq 1$.
- (43) $\rightarrow$ (44): Lemma B.8.
- (46) $\rightarrow$ (47): by the assumption, $|\{t \in D \mid t[A] = a\}| \geq 2$.

Combining the bounds for the two terms yields

$$|\mathcal{MD}(P \perp O \mid A, D', t) - \mathcal{MD}(P \perp O \mid A, D, t)| \leq \frac{1}{4n} + \frac{6}{n} \leq \frac{7}{n}.$$

□

PROOF OF LEMMA B.6 PART 2. **Unconditional case:**

Let there be $D \sim D'$ neighboring databases such that they differ in one tuple by replacement. Without loss of generality, assume that D' contains one more occurrence of $t'' = (p_{t''}, y_{t''})$ and one less occurrence of $t' = (p_{t'}, y_{t'})$ than D .

We extend $\mathcal{MD}$ to all tuples (even those not observed in D) as follows: if $t \notin D$ (i.e., the projection of t onto the attributes in F is not observed in D), then $\mathcal{MD}(F, D, t) = 0$.

By the definition, $\text{top} - k$ is a set of size k that maximizes $\sum_{t \in S} \mathcal{MD}(F, D, t)$ over all sets S such that $|S| = k$. In particular,

$$\sum_{t \in \text{top}-k} \mathcal{MD}(F, D, t) \geq \sum_{t \in \text{top}-k'} \mathcal{MD}(F, D, t)$$

, and similarly,

$$\sum_{t \in \text{top}-k'} \mathcal{MD}(F, D', t) \geq \sum_{t \in \text{top}-k} \mathcal{MD}(F, D', t).$$

This is also true for the extended definition of $\mathcal{MD}$, since an unobserved tuple in any set can be swapped by an observed one (nonnegative by the definition), thereby increasing the total $\mathcal{MD}$ of the set.

Assume without loss of generality that $\sum_{t \in \text{top}-k'} \mathcal{MD}(F, D', t) \geq \sum_{t \in \text{top}-k} \mathcal{MD}(F, D, t)$ (the other case is symmetric). We now distinguish two cases. If $\sum_{t \in \text{top}-k} \mathcal{MD}(F, D', t) \geq \sum_{t \in \text{top}-k} \mathcal{MD}(F, D, t)$, then

$$\begin{aligned} & \left| \sum_{t \in \text{top}-k'} \mathcal{MD}(F, D', t) - \sum_{t \in \text{top}-k} \mathcal{MD}(F, D, t) \right| \\ &= \sum_{t \in \text{top}-k'} \mathcal{MD}(F, D', t) - \sum_{t \in \text{top}-k} \mathcal{MD}(F, D, t) \\ &\leq \sum_{t \in \text{top}-k'} \mathcal{MD}(F, D', t) - \sum_{t \in \text{top}-k} \mathcal{MD}(F, D', t) \\ &\quad + \sum_{t \in \text{top}-k} \mathcal{MD}(F, D', t) - \sum_{t \in \text{top}-k} \mathcal{MD}(F, D, t) \\ &\leq 0 + \sum_{t \in \text{top}-k} (\mathcal{MD}(F, D', t) - \mathcal{MD}(F, D, t)) \\ &\leq \sum_{t \in \text{top}-k} |\mathcal{MD}(F, D', t) - \mathcal{MD}(F, D, t)|. \end{aligned}$$

Otherwise, $\sum_{t \in \text{top}-k} \mathcal{MD}(F, D', t) < \sum_{t \in \text{top}-k} \mathcal{MD}(F, D, t)$, and so

$$\begin{aligned} & \left| \sum_{t \in \text{top}-k'} \mathcal{MD}(F, D', t) - \sum_{t \in \text{top}-k} \mathcal{MD}(F, D, t) \right| \\ &= \sum_{t \in \text{top}-k'} \mathcal{MD}(F, D', t) - \sum_{t \in \text{top}-k} \mathcal{MD}(F, D, t) \\ &\leq \sum_{t \in \text{top}-k'} (\mathcal{MD}(F, D', t) - \mathcal{MD}(F, D, t)) \\ &\leq \sum_{t \in \text{top}-k'} |\mathcal{MD}(F, D', t) - \mathcal{MD}(F, D, t)| \end{aligned}$$

Combining the two cases yields

$$\begin{aligned} & \left| \sum_{t \in \text{top}-k'} \mathcal{MD}(F, D', t) - \sum_{t \in \text{top}-k} \mathcal{MD}(F, D, t) \right| \\ &\leq \max \left\{ \sum_{t \in \text{top}-k} |\mathcal{MD}(F, D', t) - \mathcal{MD}(F, D, t)|, \right. \\ &\quad \left. \sum_{t \in \text{top}-k'} |\mathcal{MD}(F, D', t) - \mathcal{MD}(F, D, t)| \right\} \\ &= |\mathcal{U}_{\text{TC}}(F, D') - \mathcal{U}_{\text{TC}}(F, D)| \\ &= \left| \sum_{t \in \text{top}-k'} \mathcal{MD}(F, D', t) - \sum_{t \in \text{top}-k} \mathcal{MD}(F, D, t) \right| \\ &\leq \max \left\{ \sum_{t \in \text{top}-k'} |\mathcal{MD}(F, D', t) - \mathcal{MD}(F, D, t)|, \right. \\ &\quad \left. \sum_{t \in \text{top}-k} |\mathcal{MD}(F, D', t) - \mathcal{MD}(F, D, t)| \right\} \\ &\leq k \cdot \Delta_{\mathcal{MD}(F, D, t)} \\ &\leq k \cdot \frac{3}{n} \\ &= \frac{3k}{n}, \end{aligned}$$

where $\Delta_{\mathcal{MD}(F, D, t)} \leq \frac{3}{n}$ by Lemma B.9.

Conditional case:

Similarly, assume that D' contains one more occurrence of $t'' = (p_{t''}, y_{t''}, a_{t''})$ and one less occurrence of $t' = (p_{t'}, y_{t'}, a_{t'})$ than D . Then, similarly to the unconditional case,

$$\begin{aligned} & |\mathcal{U}_{\text{TC}}(F, D') - \mathcal{U}_{\text{TC}}(F, D)| \\ &= \left| \sum_{t \in \text{top}-k'} \mathcal{MD}(F, D', t) - \sum_{t \in \text{top}-k} \mathcal{MD}(F, D, t) \right| \\ &\leq \max \left\{ \sum_{t \in \text{top}-k'} |\mathcal{MD}(F, D', t) - \mathcal{MD}(F, D, t)|, \right. \\ &\quad \left. \sum_{t \in \text{top}-k} |\mathcal{MD}(F, D', t) - \mathcal{MD}(F, D, t)| \right\} \\ &\leq k \cdot \Delta_{\mathcal{MD}(F, D, t)} \\ &\leq k \cdot \frac{7}{n} \\ &= \frac{7k}{n}, \end{aligned}$$

where $\Delta_{\mathcal{MD}(F, D, t)} \leq \frac{7}{n}$ by Lemma B.9. $\square$

The proof of Proposition 3.12 Parts 3 and 4 is similar to the respective parts of the proof of Proposition 3.4.

B.2 Proofs for Section 4

PROOF OF LEMMA 3.8. Let D_{α_i} be the set of tuples t such that $x_t = \text{True}$ under assignment α_i .

Recall that the repair cost is defined as the symmetric difference between D and D_{α_i}

$$\mathcal{U}_R(\varphi(D, D_{s_j}), \alpha_i) = |D \setminus D_{\alpha_i}| + |D_{\alpha_i} \setminus D|$$

This expression counts deletions (tuples in D but not in D_{α_i}) and insertions (tuples in D_{α_i} but not in D). Now, observe that:

- Each satisfied soft clause of the form x_t (with $t \in D$) means that t is retained in D_{α_i} (i.e., not deleted).
- Each satisfied soft clause of the form $\neg x_t$ (with $t \in D_{s_j} \setminus D$) means that t is not inserted into D_{α_i} .

Thus, satisfying more soft clauses corresponds to performing fewer changes to the original database. Specifically, more original tuples are kept, and fewer new tuples are added. Hence, the symmetric difference $|D \setminus D_{\alpha_i}| + |D_{\alpha_i} \setminus D|$ is smaller.

Therefore, if α_1 satisfies more soft clauses than α_2 , it must result in a smaller repair cost by the definition of $\mathcal{U}_R$. That is

$$\Delta(\varphi(D, D_{s_j}), \alpha_1) < \Delta(\varphi(D, D_{s_j}), \alpha_2)$$

$\square$

PROPOSITION B.10 (EXPRESSING $\mathcal{U}_R^{S\mathcal{AT}}$ IN TERMS OF DEFINITION 3.6). Let D be a database, let F be a fairness criterion, and let $\varphi(D, D_{s_j})$ be the CNF from Definition 3.6. Denote by $\mathcal{H}(D_{s_j})$ and $\mathcal{S}(D_{s_j})$ the sets of hard and soft clauses in $\varphi(D, D_{s_j})$, respectively. The following holds.

$$\mathcal{U}_R^{S\mathcal{AT}}(F, D) = |\mathcal{S}(D_{s_j})| - \min_{\alpha \models \mathcal{H}(D_{s_j})} |\{\text{soft clauses not satisfied by } \alpha\}|$$

PROOF. By Definition 3.6, each tuple $t \in D$ contributes a soft clause x_t , and each tuple $t \in D_{s_j} \setminus D$ contributes a soft clause $\neg x_t$.

For any feasible assignment α , let $D_R = \{t \in D_{s_j} \mid \alpha(x_t) = \text{True}\}$. Then, since soft clauses are a multiset, a soft clause is not satisfied iff the respective tuple appears in D and doesn't appear in D_R . That is, the number of satisfied soft clauses is

$$\#\{\text{soft clauses satisfied by } \alpha\} = |\mathcal{S}(D_{s_j})| - \Delta(\varphi(D, D_{s_j}), \alpha)$$

Therefore, minimizing $\Delta(\varphi(D, D_{sj}), \alpha)$ is equivalent to maximizing the number of satisfied soft clauses. Thus, the two definitions are also equivalent. $\square$

Following are definitions and propositions with proofs for the full $\mathcal{U}_R^{S\mathcal{AT}}$ sensitivity analysis.

Definition B.11 (Assignment extension). *Given a database D and a CNF formula $\varphi(D, D_{sj}) = \mathcal{H}(D_{sj}) \wedge \bigwedge_{t \in D} x_t \wedge \bigwedge_{t \in D_{sj}-D} (\neg x_t)$, an assignment α for $\varphi(D, D_{sj})$, and a CNF formula $\varphi' = \mathcal{H}(D_{cross}) \wedge \bigwedge_{t \in D} x_t \wedge \bigwedge_{t \in D_{cross}-D} (\neg x_t)$ such that $D_{sj} \subseteq D_{cross}$, an extension of the assignment α for $\varphi(D, D_{cross})$ is defined as follows:*

$$\alpha'(x_t) = \begin{cases} \alpha(x_t) & \text{if } t \in D_{sj} \\ \text{False} & \text{if } t \in D_{cross} \setminus D_{sj} \end{cases}$$

With this definition, we show that a minimum repair of the join and cross-product databases by their CNF formulae have identical size. We then prove that extending a CNF formula for the join database to a CNF formula for the cross-product database preserves hard clauses satisfaction.

LEMMA B.12. *Given a database D and a fairness criterion, let D_{sj} and D_{cross} be the self-join and cross-product databases. Let $\varphi(D, D_{sj})$ and $\varphi(D, D_{cross})$ be two CNF formulae defined according to Definition 3.6. Let α be an optimal assignment for $\varphi(D, D_{sj})$, and let $\hat{\alpha}$ be its extension to the assignment for $\varphi(D, D_{cross})$ according to Definition B.14. It holds that $\mathcal{U}_R^{S\mathcal{AT}}(\varphi(D, D_{sj})) = \mathcal{U}_R^{S\mathcal{AT}}(\varphi(D, D_{cross}))$.*

PROOF. Let α be an optimal assignment for $\varphi(D, D_{sj})$, and let $\hat{\alpha}$ be its extension to $\varphi(D, \hat{D})$, as defined in Definition B.14.

Since α is an optimal assignment, it satisfies all hard clauses in $\varphi(D, D_{sj})$. Therefore, by Lemma B.13, $\hat{\alpha}$ satisfies all hard clauses in $\varphi(D, \hat{D})$. Additionally, by Lemma B.15, the number of satisfied soft clauses in $\varphi(D, \hat{D})$ under $\hat{\alpha}$ is at least that in $\varphi(D, D_{sj})$ under α . Hence, $\hat{\alpha}$ is a feasible and possibly optimal assignment for $\varphi(D, \hat{D})$. Therefore, by Lemma 3.8 it holds that

$$\Delta(\varphi(D, \hat{D}), \hat{\alpha}) \leq \Delta(\varphi(D, D_{sj}), \alpha)$$

Now let $\hat{\alpha}^*$ be an optimal assignment for $\varphi(D, \hat{D})$, and let α^* be its restriction to D_{sj} defined as follows

$$\alpha^*(x_t) = \hat{\alpha}^*(x_t) \text{ for } t \in D_{sj}$$

Since $D_{sj} \subseteq \hat{D}$, then $\mathcal{H}(D_{sj}) \subseteq \mathcal{H}(\hat{D})$. And because hard clause satisfaction is preserved under restriction (Lemma B.13), it follows that α^* is a feasible assignment for $\varphi(D, D_{sj})$.

Moreover, the reverse direction of Lemma B.15 implies that the number of soft clauses satisfied under α^* is at least that in $\varphi(D, D_{sj})$ under any other assignment. This means that α^* is an optimal assignment α for $\varphi(D, D_{sj})$, and $\hat{\alpha}$ is an optimal assignment $\hat{\alpha}^*$ for $\varphi(\hat{D})$ by Definition B.14.

Therefore, by Lemma 3.8 it holds that

$$\Delta(\varphi(D, D_{sj}), \alpha) = \Delta(\varphi(D, D_{sj}), \alpha^*) \leq \Delta(\varphi(D, \hat{D}), \hat{\alpha}) = \Delta(\varphi(D, \hat{D}), \hat{\alpha}^*)$$

So from Definition 3.9 and from both directions we proved, it holds that

$$\begin{aligned} \mathcal{U}_R^{S\mathcal{AT}}(F, D) &= \Delta(\varphi(D, D_{sj}), \alpha) \\ &= \Delta(\varphi(D, \hat{D}), \hat{\alpha}) = \min_{\alpha \in \mathcal{H}(\hat{D})} \Delta(\varphi(D, \hat{D}), \alpha) \end{aligned}$$

$\square$

We will now prove that extending a CNF formula for the join database to a CNF formula for the cross-product database preserves hard clauses satisfaction.

LEMMA B.13. *Given a database D such that the schema of D is a superset of $P \cup O \cup A$, and a fairness criterion of the form $F = P \perp O \mid A$, let D_{sj} be the join database and D_{cross} be cross-product database, and let $\varphi(D, D_{sj})$ and $\varphi(D, D_{cross})$ be the corresponding two CNF formulae from Definition 3.6. Let α be a feasible assignment for $\varphi(D, D_{sj})$ and let $\hat{\alpha}$ be its extension to the assignment for $\varphi(D, D_{cross})$ according to Definition B.14. Then, $\mathcal{H}(D_{sj})$ in $\varphi(D, D_{sj})$ are satisfied by α iff $\mathcal{H}(\hat{D}_{sj})$ in $\varphi(D, D_{cross})$ are satisfied by $\hat{\alpha}$.*

PROOF. ($\Rightarrow$) Suppose that $\mathcal{H}(D_{sj})$ in $\varphi(D, D_{sj})$ are satisfied under α . Since $\hat{D}$ is a superset of D_{sj} , then each clause in $\mathcal{H}(D_{sj})$ appears in $\mathcal{H}(\hat{D}_{sj})$, and they differ only in additional hard clauses added because of the tuples in $\hat{D} \setminus D_{sj}$. Since $\hat{\alpha}$ agrees with α on all $t \in D_{sj}$, the same clauses also evaluate to True under $\hat{\alpha}$.

Now consider the additional clauses introduced due to tuples in $\hat{D} \setminus D_{sj}$. These clauses are of the form $(\neg x_{t_1} \vee \neg x_{t_2} \vee x_{t_3})$, where at least one of the tuples t_1, t_2, t_3 is in $\hat{D} \setminus D_{sj}$. Since $\hat{\alpha}$ assigns False to all x_t so that $t \in \hat{D} \setminus D_{sj}$, the clause $(\neg x_{t_1} \vee \neg x_{t_2} \vee x_{t_3})$ will evaluate to True as follows

- If either x_{t_1} or x_{t_2} is False under $\hat{\alpha}$, their negation is True, so the whole clause is satisfied by $\hat{\alpha}$.
- If both x_{t_1} and x_{t_2} are True under $\hat{\alpha}$, this is possible only if both t_1 and t_2 are in D_{sj} .

According to the construction of the CNF formula (see Definition 3.6), a clause of the form

$$(\neg x_{t_1} \vee \neg x_{t_2} \vee x_{t_3})$$

is included in the hard clauses, where $t_1 = (P_1, O_1, A) \in D_{sj}$, $t_2 = (P_2, O_2, A) \in D_{sj}$, and $t_3 = (P_1, O_2, A)$ is the tuple resulting from the join of t_1 and t_2 on the shared attribute A .

Since both t_1 and t_2 belong to D_{sj} , and the join $D_{sj}(P_1, O_2, A) := D(P_1, O_1, A) \bowtie D(P_2, O_2, A)$ preserves A , the resulting tuple $t_3 = (P_1, O_2, A)$ must also belong to D_{sj} . Therefore, the clause $(\neg x_{t_1} \vee \neg x_{t_2} \vee x_{t_3})$ belongs to $\mathcal{H}(D_{sj})$, and since $\hat{\alpha}$ agrees with α on all tuples in D_{sj} , it satisfies this clause as well. Then the full clause is in $\mathcal{H}(D_{sj})$ and should be satisfied by $\hat{\alpha}$ because it agrees with α on all $t \in D_{sj}$.

Therefore, in any case $\mathcal{H}(\hat{D}_{sj})$ in $\varphi(D, \hat{D})$ are satisfied under $\hat{\alpha}$.

($\Leftarrow$) Suppose $\mathcal{H}(\hat{D}_{sj})$ in $\varphi(D, \hat{D})$ are satisfied under $\hat{\alpha}$. In particular, all clauses in $\mathcal{H}(D_{sj})$, which are a subset of $\mathcal{H}(\hat{D}_{sj})$, are satisfied under $\hat{\alpha}$. Since $\hat{\alpha}$ agrees with α on all x_t for $t \in D_{sj}$, then $\mathcal{H}(D_{sj})$ in $\varphi(D, D_{sj})$ are satisfied also under α .

From the two directions, it holds that $\mathcal{H}(D_{sj})$ in $\varphi(D, D_{sj})$ are satisfied under α iff $\mathcal{H}(\hat{D}_{sj})$ in $\varphi(D, \hat{D})$ are satisfied under $\hat{\alpha}$. $\square$

Finally, we show that extending a CNF formula for the self-join database to a CNF formula for the cross-product database cannot decrease the number of satisfied soft clauses.

Definition B.14 (Assignment extension). *Given a database D and a CNF formula $\varphi(D, D_{sj}) = \mathcal{H}(D_{sj}) \wedge \bigwedge_{t \in D} x_t \wedge \bigwedge_{t \in D_{sj}-D} (\neg x_t)$, an assignment α for $\varphi(D, D_{sj})$, and a CNF formula $\varphi' = \mathcal{H}(D_{cross}) \wedge$*

$\bigwedge_{t \in D} x_t \wedge \bigwedge_{t \in D_{\text{cross}} - D} (\neg x_t)$ such that $D_{sj} \subseteq D_{\text{cross}}$, an extension of the assignment α for $\varphi(D, D_{\text{cross}})$ is defined as follows:

$$\alpha'(x_t) = \begin{cases} \alpha(x_t) & \text{if } t \in D_{sj} \\ \text{False} & \text{if } t \in D_{\text{cross}} \setminus D_{sj} \end{cases}$$

LEMMA B.15. *Given a database D and a fairness criterion, let D_{sj} and D_{cross} be the join and cross-product databases. Let $\varphi(D, D_{sj})$ and $\varphi(D, D_{\text{cross}})$ be the two corresponding CNF formulae according to Definition 3.6. Let α be a feasible assignment for $\varphi(D, D_{sj})$ and let $\hat{\alpha}$ be its extension to the assignment for $\varphi(D, D_{\text{cross}})$. Then the number of satisfied soft clauses in $\varphi(D, D_{sj})$ by α is lower than or equal to the number of satisfied soft clauses in $\varphi(D, D_{\text{cross}})$ under $\hat{\alpha}$.*

PROOF. Let D be a dataset. According to Definition 3.6:

$$\begin{aligned} \varphi(D, D_{sj}) &= \mathcal{H}(D_{sj}) \wedge \bigwedge_{t \in D} x_t \wedge \bigwedge_{t \in D_{sj} \setminus D} \neg x_t \\ \varphi(D, \hat{D}) &= \mathcal{H}(\hat{D}) \wedge \bigwedge_{t \in D} x_t \wedge \bigwedge_{t \in \hat{D} \setminus D} \neg x_t \end{aligned}$$

Since $D_{sj} \subseteq \hat{D}$, then $\varphi(D, D_{sj})$ and $\varphi(D, \hat{D})$ differ only in soft clauses for the tuples in $\hat{D} \setminus D_{sj}$. In particular, the only difference is that $\varphi(D, \hat{D})$ contains additional soft clauses of the form $\neg x_t$ for $t \in \hat{D} \setminus D_{sj}$.

Let α be an assignment for $\varphi(D, D_{sj})$. According to Definition B.14, it holds that

- All soft clauses in $\varphi(D, D_{sj})$ are also present in $\varphi(D, \hat{D})$ and are satisfied under $\hat{\alpha}$ exactly as in α .
- The additional soft clauses in $\varphi(D, \hat{D})$ are of the form $\neg x_t$ for $t \in \hat{D} \setminus D_{sj}$, and each such clause is satisfied in $\hat{\alpha}$ because $\hat{\alpha}(x_t) = \text{False}$.

Hence, $\hat{\alpha}$ satisfies all the soft clauses that α satisfies, plus potentially some new ones. Therefore, the number of satisfied soft clauses in $\varphi(D, D_{sj})$ under α is less or equal to the number of satisfied soft clauses in $\varphi(D, \hat{D})$ under $\hat{\alpha}$. $\square$

As a precursor to the sensitivity analysis, we show that the difference in repair between neighboring databases is bounded by 2.

LEMMA B.16. *Let $D \sim D'$, let D_{cross} and D'_{cross} be their corresponding cross-product databases, and let $F = P \perp\!\!\!\perp O \mid A$ be a fairness criterion. Let $\varphi(D, D_{\text{cross}})$ and $\varphi(D', D'_{\text{cross}})$ be CNF formulae defined according to Definition 3.6, and let $\hat{\alpha}$ be a feasible assignment for $\varphi(D, D_{\text{cross}})$. Finally, let $\Delta(\varphi(D, D_{\text{cross}}), \hat{\alpha})$ and $\Delta(\varphi(D', D'_{\text{cross}}), \hat{\alpha})$ be defined according to Definition 3.7. It holds that:*

$$|\Delta(\varphi(D, D_{\text{cross}}), \hat{\alpha}) - \Delta(\varphi(D', D'_{\text{cross}}), \hat{\alpha})| \leq 2$$

PROOF. Let there be $D \sim D'$ neighboring databases such that they differ in one tuple by replacement. Without loss of generality, assume that D' contains one more occurrence of t'' and one less occurrence of t' than D .

It follows that the only difference between $\varphi(D, \hat{D})$ and $\varphi(D', \hat{D}')$ lies at most in two soft clauses:

- If $t' \notin D'$, then $\varphi(D', \hat{D}')$ contains the soft clause $\neg x_{t'}$, while $\varphi(D, \hat{D})$ contains the soft clause $x_{t'}$.
- If $t'' \notin D$, then $\varphi(D', \hat{D}')$ contains the soft clause $x_{t''}$, while $\varphi(D, \hat{D})$ contains the soft clause $\neg x_{t''}$.

The set of hard clauses remains the same, as they are defined over $\hat{D}$, which depends only on the attribute domains. And since there is no difference in the hard clauses between D and D' , then $\hat{\alpha}$ is also a feasible assignment for $\varphi(D', \hat{D}')$.

Now, observe that under $\hat{\alpha}$

- If $\hat{\alpha}(x_t) = \text{True}$, then $\varphi(D', \hat{D}')$ gains one satisfied soft clause (x_t), and $\varphi(D, \hat{D})$ loses one satisfied soft clause ($\neg x_t$ violated).
- If $\hat{\alpha}(x_t) = \text{False}$, then $\varphi(D', \hat{D}')$ loses one soft clause and $\varphi(D, \hat{D})$ gains one.

Thus, the number of satisfied soft clauses under $\hat{\alpha}$ differs by at most 2 between the two formulas. As a result, the sets of tuples $D_{\hat{\alpha}}$ (tuples with $x_t = \text{True}$) used to compute the repair values may differ by at most one tuple, and the database D differs from D' by one tuple as well. Therefore, the symmetric difference

$$|D \setminus D_{\hat{\alpha}}| + |D_{\hat{\alpha}} \setminus D| \quad \text{vs} \quad |D' \setminus D_{\hat{\alpha}}| + |D_{\hat{\alpha}} \setminus D'|$$

can differ by at most 2.

Hence

$$|\Delta(\varphi(D, \hat{D}), \hat{\alpha}) - \Delta(\varphi(D', \hat{D}'), \hat{\alpha})| \leq 2$$

$\square$

We are now ready to combine the lemmas into our result that bounds the sensitivity of $\mathcal{U}_R^{S\mathcal{AT}}$.

PROPOSITION B.17 (SENSITIVITY OF $\mathcal{U}_R^{S\mathcal{AT}}$). *For a database D and a fairness criterion F , the sensitivity of $\mathcal{U}_R^{S\mathcal{AT}}(F, D)$ is at most 2.*

PROOF OF PROPOSITION B.17. Let there be $D \sim D'$ neighboring databases such that they differ in one tuple by replacement. Without loss of generality, assume that D' contains one more occurrence of t'' and one less occurrence of t' than D .

Given a fairness criterion of the form $F = P \perp\!\!\!\perp O \mid A$, let D_{sj} and $\hat{D}$ be the join and cross-product databases, and similarly for D' .

Since $\hat{D} = \hat{D}'$, (the cross-product database depends only on domains), the set of hard clauses is identical in both formulas $\varphi(D, \hat{D})$ and $\varphi(D', \hat{D}')$. We can also notice that the only difference between these formulas lies in at most two soft clauses, as formulated in the proof of Lemma B.16.

By Lemma B.12,

$$\mathcal{U}_R^{S\mathcal{AT}}(F, D) = \min_{\alpha \models \mathcal{H}(\hat{D})} \Delta(\varphi(D, \hat{D}), \alpha), \quad \mathcal{U}_R^{S\mathcal{AT}}(F, D') = \min_{\alpha \models \mathcal{H}(\hat{D})} \Delta(\varphi(D', \hat{D}'), \alpha)$$

Let

$$\alpha_D \in \arg \min_{\alpha \models \mathcal{H}(\hat{D})} \Delta(\varphi(D, \hat{D}), \alpha)$$

and

$$\alpha_{D'} \in \arg \min_{\alpha \models \mathcal{H}(\hat{D})} \Delta(\varphi(D', \hat{D}'), \alpha)$$

be optimal assignments.

By Lemma B.16, for every feasible α , it holds that

$$|\Delta(\varphi(D, \hat{D}), \alpha) - \Delta(\varphi(D', \hat{D}'), \alpha)| \leq 2$$

Applying this bound with $\alpha = \alpha_D$ and using optimality of the assignments, we get

$$\begin{aligned} \mathcal{U}_R^{S\mathcal{AT}}(F, D) &= \Delta(\varphi(D, \hat{D}), \alpha_D) \\ &\leq \Delta(\varphi(D', \hat{D}'), \alpha_D) + 2 \\ &\leq \Delta(\varphi(D', \hat{D}'), \alpha_{D'}) + 2 \\ &= \mathcal{U}_R^{S\mathcal{AT}}(F, D') + 2 \end{aligned}$$

Table 5: Results of exploratory queries q_1 – q_6 on the Adult and Compas datasets.

(a) Adult dataset		(b) Compas dataset	
q_1 : average income>50K by sex		q_4 : average decile_score by race (grouped by age_cat)	
sex	avg-i	age_cat	avg-d
Female	0.109	25 - 45	3.766
Male	0.304	Greater than 45	2.636
q_2 : average income>50K by race		Less than 25	5.201
race	avg-i	q_5 : median decile_score by race (grouped by c_charge_degree)	
Amer-Indian-Eskimo	0.117	race	med-d
Asian-Pac-Islander	0.269	African-American	4
Black	0.121	Asian	2
Other	0.123	Caucasian	2
White	0.254	Hispanic	2
		Native American	4
		Other	1
		q_6 : median decile_score by c_charge_degree (grouped by race)	
		c_charge_degree	med-d
		F	3.5
		M	2
		O	2.25

By symmetry (if we swap D and D'), we also have $\mathcal{U}_R^{SAT}(F, D') \leq \mathcal{U}_R^{SAT}(F, D) + 2$. Therefore, we get

$$\Delta_{\mathcal{U}_R^{SAT}(F,D)} = \max_{D' \sim D} |\mathcal{U}_R^{SAT}(F, D) - \mathcal{U}_R^{SAT}(F, D')| \leq 2$$

□

PROOF OF PROPOSITION 3.10 PART 4. According to Proposition B.17, for a single fairness criterion it holds that the difference between the soft clauses in the CNFs for two neighboring databases is at most one, and also that the difference in the cost of repair is at most 2. So for a set of fairness criteria $\mathcal{F}$ it holds that the difference between the soft clauses in the CNFs is at most $|\mathcal{F}|$.

By repeating the proof of Proposition B.17, we can obtain that the difference in the cost of repair between two neighboring databases for a set of fairness criteria $\mathcal{F}$ is at most $2|\mathcal{F}|$. □

C COMPLEXITY ANALYSIS OF THE ALGORITHMS

Complexity of Algorithm 1. The complexity of Algorithm 1 is $O(|\mathcal{F}|n)$. For every F , the algorithm first computes the joint and the marginal empirical probabilities. Assuming the counts are stored in hash tables, so every lookup and update takes $O(1)$ time, this whole part takes $O(n)$ time. Then, the algorithm computes the TVD and divides into two cases. In the unconditional case it goes over all the tuples in D once when calculating the sum, so the whole computation takes $O(n)$ time. And in the conditional case it goes over all the values a in A and calculates the sum from the unconditional case for it. This computation takes $O(n + n) = O(n)$ time. Updating the cumulating sum takes $O(1)$ time. Finally, after the algorithm finishes going over all F in $\mathcal{F}$, it adds Laplace noise to the resulting cumulative sum and returns it, which takes $O(1)$.

Complexity of Algorithm 2. The complexity of Algorithm 2 is $O(|\mathcal{F}|(n^4 + SAT))$, where $O(SAT)$ is the complexity of the SAT solver, due to the following analysis. For every F , the algorithm first computes the self-join D_{sj} of D on A and goes over it to add the soft

clauses to φ . It takes $O(n)$ time to calculate the per- a projection sets, so it takes $O(n^2)$ overall to perform the join and iterate over it to compute the soft clauses. Then the algorithm computes the self-join C of D_{sj} on A . Since each copy of D_{sj} contains $O(|P_a| \times |O_a|)$ tuples for any a , this takes $O(\sum_{a \in A} |P_a|^2 |O_a|^2) = O(n^4)$ time. Finally, the algorithm runs a solver to get an assignment for the constructed φ , computes the symmetric difference and updates the cumulative sum. This whole part takes $O(SAT + n)$. After the algorithm finishes iterating over all $F \in \mathcal{F}$, it adds Laplace noise to the resulting sum and returns it, which takes $O(1)$ time.

Complexity of Algorithm 3. The complexity of Algorithm 3 is $O(|\mathcal{F}|n \log n)$. For every F , the algorithm first computes the joint and the marginal empirical probabilities, which takes $O(n)$ time. Then, the algorithm computes the marginal differences, taking $O(n)$ time. Sorting of the marginal differences for all the tuples and finding the k largest ones takes $O(n \log k)$ time if we are using a size- k heap. Summing the k largest marginal differences takes $O(k)$ time and updating the cumulative sum takes $O(1)$ time. Therefore, this whole part takes $O(n \log k + k) = O(n \log n)$. After going over all $F \in \mathcal{F}$, it adds Laplace noise to the cumulative sum and returns it, which takes $O(1)$ time.

C.1 Case Study: Query Results

The results of the exploratory queries in Table 6, shown in Table 5, reveal clear disparities across protected groups in both Adult and Compas datasets.

C.2 Measure Drill-Down

Proxy faithfulness of $\mathcal{U}_{MI}^{TVD}$. Following Figure 6, Figure 9 shows a comparison of $\mathcal{U}_{MI}$ and $\mathcal{U}_{MI}^{TVD}$ values for every criterion from Table 2, with privacy budget $\varepsilon = 1$. For every dataset, $\mathcal{U}_{MI}^{TVD}$ closely tracks $\mathcal{U}_{MI}$, with an average Kendall's tau correlation [52] being

Table 6: Exploratory queries on the Adult and Compas datasets.

Dataset	Query
Adult	q_1 : SELECT sex, AVG(income>50K) AS avg-i FROM Adult GROUP BY sex;
	q_2 : SELECT race, AVG(income>50K) AS avg-i FROM Adult GROUP BY race;
	q_3 : SELECT age_cat, AVG(avg-d) AS avg-d FROM (SELECT race, age_cat, AVG(decile_score) AS avg-d FROM Compas GROUP BY race, age_cat) t GROUP BY age_cat;
	q_4 : SELECT race, AVG(avg-d) AS avg-d FROM (SELECT age_cat, race, AVG(decile_score) AS avg-d FROM Compas GROUP BY age_cat, race) t GROUP BY race;
Compas	q_5 : SELECT race, PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY med-d) AS med-d FROM (SELECT c_charge_degree, race, PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY decile_score) AS med-d FROM Compas GROUP BY c_charge_degree, race) t GROUP BY race;
	q_6 : SELECT c_charge_degree, PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY med-d) AS med-d FROM (SELECT race, c_charge_degree, PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY decile_score) AS med-d FROM Compas GROUP BY race, c_charge_degree) t GROUP BY c_charge_degree;

0.92. Specifically, for each individual dataset, an increase in $\mathcal{U}_{MI}$ corresponds to an increase in $\mathcal{U}_{MI}^{TVD}$, and vice versa, as we would expect from a good proxy. However, this proportionality does not extend across datasets, since for the Healthcare and Compas datasets the same value of $\mathcal{U}_{MI}$ (0.09) corresponds to different values of $\mathcal{U}_{MI}^{TVD}$. Within each dataset, we can further glean the level of unfairness for a criterion by comparing the values of the same measure for it and for another criterion. For example, for IPUMS-CPS, both measures assign higher values to criterion 4 (HEALTH $\perp$ INCTOT | AGE) than to criterion 3 (HEALTH $\perp$ MARST | AGE), meaning that the predicted unfairness for criterion 4 is larger. Intuitively, given a person’s age, their income depends more on their health than on marital status.

Effect of k on $\mathcal{U}_{TC}$. Figure 10 depicts the effect of the parameter k on the value and relative $L1$ error of Algorithm 3 (computing $\mathcal{U}_{TC}$). In Figure 10a, we plot the value of Algorithm 3 as k increases, with infinite privacy budget, $\epsilon = \infty$. We do so since this experiment examines the effect of k on the measure values, which would be highly distorted by noise, due to the dependency of the noise scale on k (see Proposition 3.12). Overall, as expected, the value is monotone non-decreasing in k because, as k grows, we add an increasing number of non-negative $\mathcal{MD}$ values to the resulting sum. Almost all datasets reach a ‘plateau’ starting from some value of k , meaning that a subset of tuples carries most of the $\mathcal{MD}$ of the dataset, while the remaining tuples contribute only marginally as k increases. For example, in Healthcare, most tuples have large $\mathcal{MD}$ values, and thus $\mathcal{U}_{TC}$ continues to grow almost linearly in k . In contrast, for IPUMS-CPS, many tuples have small $\mathcal{MD}$ values, so the trend does not plateau and remains noisy.

Figure 10b depicts the relative $L1$ error as a function of k , due to noise. The relative error is quite large due the sensitivity of $\mathcal{U}_{TC}$ being directly proportional to k and $|\mathcal{F}|$, and thus also the magnitude of the added DP noise (Proposition 3.12). Furthermore, the error is larger and exhibits greater variability for datasets with many distinct values, such as IPUMS-CPS. This suggests that a larger budget may be required for $\mathcal{U}_{TC}$ and that it may be used as a secondary measure for identifying the cumulative effect of outlier tuples. We recognize that selecting an optimal value of k for $\mathcal{U}_{TC}$ is a challenging problem and leave this for future work.

Comparing $\mathcal{U}_{MI}^{TVD}$ and $\mathcal{U}_{TC}$. In Figure 8, we compared $\mathcal{U}_{MI}^{TVD}$ and $\mathcal{U}_{TC}$ without privacy considerations. While the values of $\mathcal{U}_{MI}^{TVD}$ are smaller than the values of $\mathcal{U}_{TC}$ for all criteria, it can be noticed that $\mathcal{U}_{MI}^{TVD}$ and $\mathcal{U}_{TC}$ exhibited exactly the same trends in values. That is, higher values of $\mathcal{U}_{MI}^{TVD}$ correspond to higher values of $\mathcal{U}_{TC}$, and vice versa. Moreover, relative differences in the values of $\mathcal{U}_{MI}^{TVD}$ are reflected by comparable relative differences in the values of $\mathcal{U}_{TC}$.

Chunk heuristic effect on $\mathcal{U}_R^{SAT}$. In Figure 7, we compared Algorithm 2 with and without the heuristic (i.e., computing $\mathcal{U}_R^{SAT}$ with chunks of size 100, and without them), with a budget of $\epsilon = 1$. Algorithm 2 with the heuristic is noticeably faster than without it, reducing the total runtime by 56.26% across all datasets. The values for the heuristic almost exactly mirror the trend of the values for the true $\mathcal{U}_R^{SAT}$.

For the rest of this section we will denote $\mathcal{U}_R^{SAT}$ with the chunking heuristic as $\mathcal{U}_R^{SAT_{chunk}}$. We will prove that the sensitivity bound of $\mathcal{U}_R^{SAT_{chunk}}$ is the same as the sensitivity bound of $\mathcal{U}_R^{SAT}$.

PROPOSITION C.1 (SENSITIVITY OF $\mathcal{U}_R^{SAT_{chunk}}$). *For a database D and a fairness criterion F , the sensitivity of $\mathcal{U}_R^{SAT_{chunk}}(F, D)$ is at most 2.*

PROOF. Let there be $D \sim D'$ neighboring databases such that they differ in one tuple by replacement. Let $\text{Chunk}(\cdot)$ be the chunking procedure used by $\mathcal{U}_R^{SAT_{chunk}}$, which consists of sorting and then partitioning into consecutive blocks of fixed size, such that we assume that all duplicates of any tuple are placed into the same block, the blocks are disjoint and their union equals the full database. We write

$$\text{Chunk}(D) = \{D^{(1)}, \dots, D^{(m)}\}, \quad \text{Chunk}(D') = \{D'^{(1)}, \dots, D'^{(m)}\}$$

By the definition of $\mathcal{U}_R^{SAT_{chunk}}$, we assume that any neighbor D' of D has the same tuple ordering as D . Therefore, the replacement of a single tuple can affect the contents of at most one chunk, which is the corresponding chunk in D' . Hence, there exists an index $j \in \{1, \dots, m\}$ such that

$$D^{(i)} = D'^{(i)} \text{ for all } i \neq j, \quad D^{(j)} \sim D'^{(j)}$$

By the definition of $\mathcal{U}_R^{SAT_{chunk}}$, it holds that

$$\mathcal{U}_R^{SAT_{chunk}}(F, D) = \sum_{i=1}^m \mathcal{U}_R^{SAT}(F, D^{(i)}), \quad \mathcal{U}_R^{SAT_{chunk}}(F, D') = \sum_{i=1}^m \mathcal{U}_R^{SAT}(F, D'^{(i)})$$

Therefore,

$$\begin{aligned} \left| \mathcal{U}_R^{SAT_{chunk}}(F, D) - \mathcal{U}_R^{SAT_{chunk}}(F, D') \right| &= \left| \sum_{i=1}^m \mathcal{U}_R^{SAT}(F, D^{(i)}) - \sum_{i=1}^m \mathcal{U}_R^{SAT}(F, D'^{(i)}) \right| \\ &= \left| \mathcal{U}_R^{SAT}(F, D^{(j)}) - \mathcal{U}_R^{SAT}(F, D'^{(j)}) \right| \end{aligned}$$

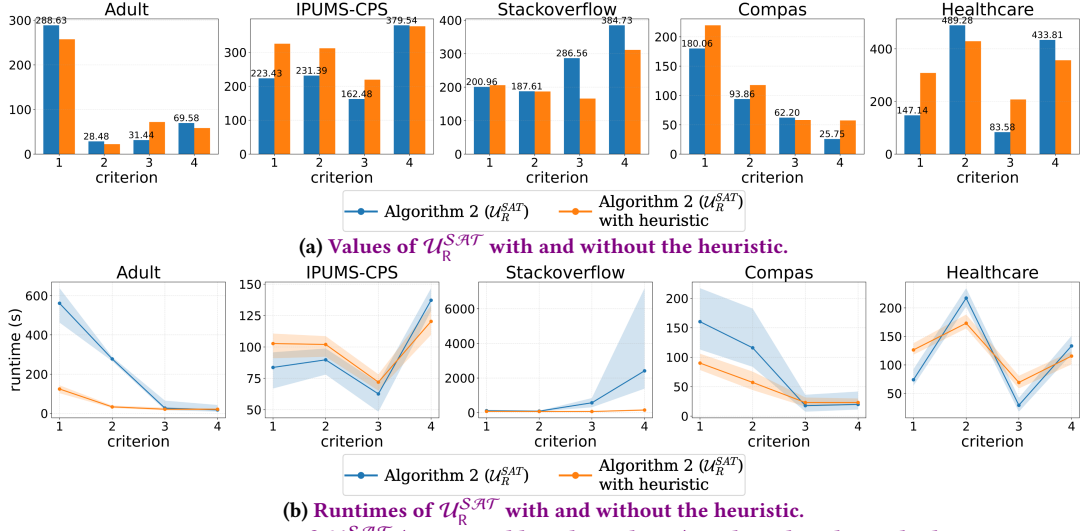


Figure 7: Comparison of $\mathcal{U}_R^{SAT}$ (computed by Algorithm 2) with and without the heuristic.

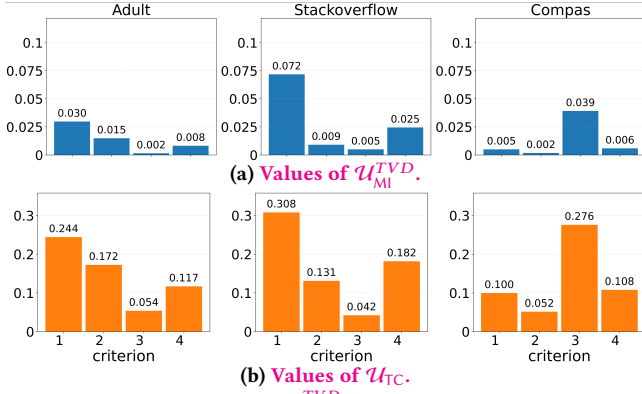


Figure 8: Comparison of $\mathcal{U}_{MI}^{TVD}$ (computed by Algorithm 1) and $\mathcal{U}_{TC}$ (computed by Algorithm 3) without privacy considerations.

Since $D^{(j)} \sim D'^{(j)}$, by Proposition B.17) we get

$$\left| \mathcal{U}_R^{SAT}(F, D^{(j)}) - \mathcal{U}_R^{SAT}(F, D'^{(j)}) \right| \leq 2.$$

Combining the above yields

$$\left| \mathcal{U}_R^{SAT_{\text{chunk}}}(F, D) - \mathcal{U}_R^{SAT_{\text{chunk}}}(F, D') \right| \leq 2$$

Therefore, we get

$$\Delta_{\mathcal{U}_R^{SAT_{\text{chunk}}}(F, D)} = \max_{D' \sim D} \left| \mathcal{U}_R^{SAT_{\text{chunk}}}(F, D) - \mathcal{U}_R^{SAT_{\text{chunk}}}(F, D') \right| \leq 2$$

□

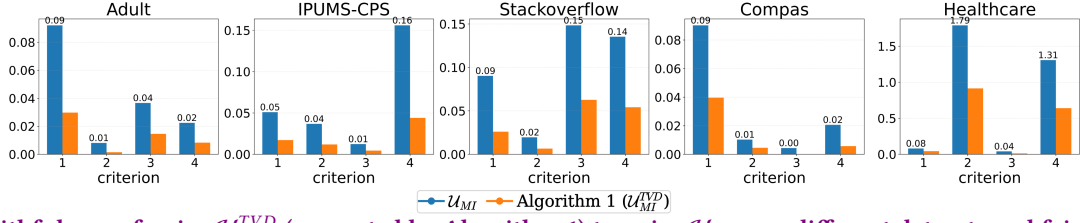


Figure 9: Faithfulness of noisy $\mathcal{U}_{MI}^{TVD}$ (computed by Algorithm 1) to noisy $\mathcal{U}_{MI}$ over different datasets and fairness criteria from Table 2.

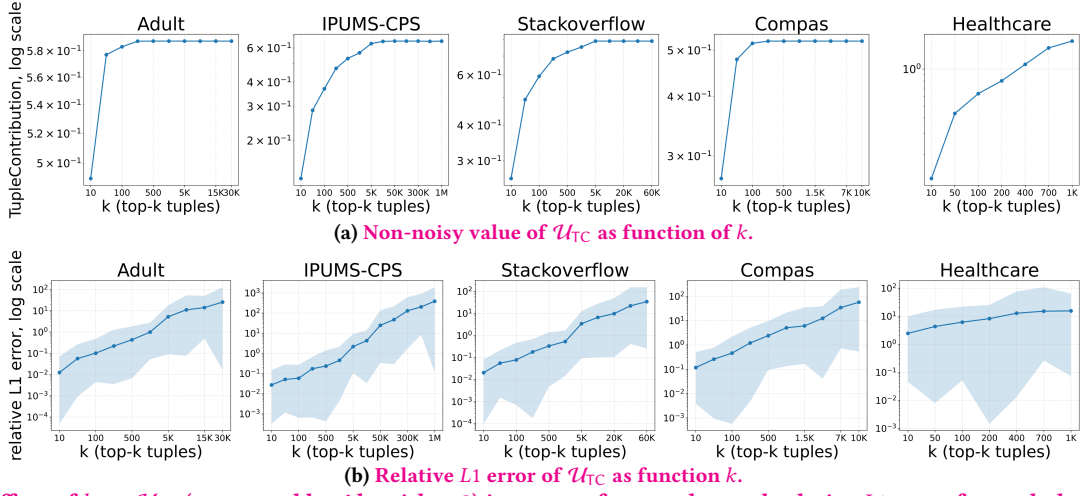


Figure 10: Effect of k on $\mathcal{U}_{TC}$ (computed by Algorithm 3) in terms of true value and relative $L1$ error for each dataset with its four criteria.